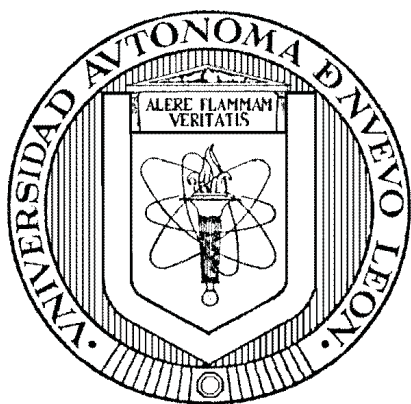


UNIVERSIDAD AUTÓNOMA DE NUEVO LEÓN
FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA
SUBDIRECCIÓN DE ESTUDIOS DE POSGRADO



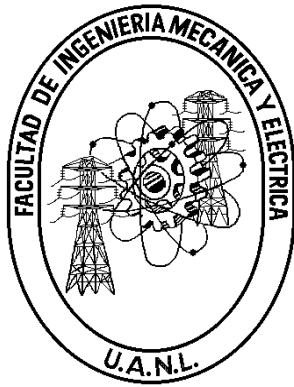
YIELD-AWARE HIGHER-ORDER TRAJECTORY OPTIMIZATION FOR
STEP-AND-SCAN WAFER STAGES

POR
ROBERTO TREVIÑO CERVANTES

COMO REQUISITO PARA OBTENER EL GRADO DE
MAESTRÍA EN CIENCIAS DE LA INGENIERÍA ELÉCTRICA

OCTUBRE 2026

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Los miembros del Comité de Tesis recomendamos que la tesis “**Yield-Aware Higher-Order Trajectory Optimization for Step-and-Scan Wafer Stages**” realizada por el(la) alumno(a) **Roberto Treviño Cervantes**, con número de matrícula 1915003, sea aceptada para su defensa como opción al grado de **Maestría en Ciencias de la Ingeniería Eléctrica**.

El Comité de Tesis

Dr. Juan Angel Rodriguez Liñan
Director

Dr. Nombre Del Revisor Interno
Revisor Interno

Dr. Nombre Del Revisor Externo
Revisor Externo

Vo.Bo.

Dr. Simon Martinez Martinez
Subdirector de Estudios de Posgrado

San Nicolás de los Garza, Nuevo León, México. Octubre 2026

A ... por su apoyo ... ETC

Agradecimientos

Agradezco a ... blah blah blah.

Abstract

Step-and-scan photolithography scanners expose the layers of an integrated circuit by sweeping a reticle and a silicon wafer through a projection slit in opposite directions, synchronized to nanometre-level accuracy. The reference trajectories that command these motions determine how much energy is injected into the poorly damped structural modes of the machine, and therefore how large the residual synchronization error is during exposure. Trajectory design is conventionally evaluated through servo-mechanical metrics — settling time, tracking error, and the moving average (MA) and moving standard deviation (MSD) of the synchronization error — none of which describes the morphology of the defects that those errors ultimately imprint on the wafer.

This work proposes a yield-aware framework for optimizing the kinematic bounds ($v_{\max}$, $a_{\max}$, $j_{\max}$, $s_{\max}$) of fourth-order (snap-limited) reference trajectories. A planar multi-body model of a generic lithography machine, comprising the reticle, optics, and wafer chains coupled through Kelvin–Voigt elements, is implemented as a simulation in the MuJoCo physics engine and validated against the characterization study it derives from. The synchronization error produced during scanning is transformed into a simulated wafer defect map, which a convolutional neural network (CNN) trained on the industrial WM-811K dataset (172,950 labelled wafer maps, 96.88 % hold-out accuracy) classifies into nine defect morphologies. The probability distribution estimated

by the CNN, restricted to the classes plausibly caused by scanner motion, defines a cost function that links trajectory parameters directly to expected yield, closing the loop for data-driven trajectory optimization under a throughput constraint.

Keywords: photolithography, step-and-scan, trajectory generation, fourth-order profiles, multi-body dynamics, convolutional neural networks, wafer maps, yield.

Symbols and notation

$\dot{x}, \ddot{x}$	Dots denote derivatives with respect to time, e.g. $\dot{x} = dx/dt$.
x^T, A^T	Transpose of the vector x and the matrix A .
$\mathbb{R}$	The field of real numbers.
$\ x\ $	A norm of x .
$\mathcal{F}_0, \mathcal{F}_1$	World (inertial) frame and base frame of the machine.
j	Chain index: $j = 1$ reticle, $j = 2$ optics, $j = 3$ wafer.
i_j	Body i within chain j ; N_j is the last body of chain j .
f_{i_j}	In-plane displacement of body i_j relative to its parent body.
$\theta_{z_{i_j}}$	Rotation of body i_j about the vertical axis.
$m_{i_j}, I_{z_{i_j}}$	Mass and z -axis moment of inertia of body i_j .
K_{i_j}, C_{i_j}	Stiffness and damping of the Kelvin–Voigt coupling of body i_j .
α	Projection demagnification factor, $\alpha = 0.25$.
e_{syn}	Synchronization error between wafer and reticle stages.
MA, MSD	Moving average and moving standard deviation of e_{syn} over the exposure window.
$v_{\max}, a_{\max}$	Velocity and acceleration bounds of the reference trajectory.
$\dot{j}_{\max}, s_{\max}$	Jerk and snap bounds of the reference trajectory.
CD	Critical dimension: minimum printable feature size.
NA	Numerical aperture of the projection lens.
λ	Illumination wavelength.
P_c	Probability assigned by the CNN to defect class c .
w_c	Economic weight of defect class c in the cost function.
$\mathcal{J}$	Trajectory cost: expected yield loss, $\mathcal{J} = \sum_c w_c P_c$.
WPH	Throughput in wafers per hour.
<i>Italics</i>	Italics denote terms in a language other than English.
Boldface	Boldface denotes technical terms at first use.

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Chapter 1

Introduction

The history of modern electronics is inseparable from the history of lithography. Since the independent invention of the integrated circuit (IC) by Jack Kilby at Texas Instruments and Robert Noyce at Fairchild Semiconductor in 1958, Moore’s Law has governed the relentless progression of the semiconductor industry: the density of components on a chip doubles approximately every two years¹³. This exponential trend has translated into dramatic reductions in cost alongside equally dramatic increases in performance. The price of one gigabyte of NAND flash memory, for instance, fell from over \$1000 at the start of this millennium to below \$0.30 by the mid-2010s³, while the minimum transistor feature size has shrunk from hundreds of nanometres to just a few tens of nanometres. The engine of this revolution is *photolithography* — the process by which the intricate patterns of a transistor, a wire, or a memory cell are projected onto a silicon wafer with nanometre-scale fidelity.

The fabrication of a modern IC consists of up to 30 sequential process cycles for mature technology nodes¹³, each defining one distinct layer of the device; at leading-edge nodes — such as 5 nm CMOS production technology — the layer count rises to approximately 75^{17;12}. A monocrystalline silicon wafer — a thin disc of up to 300 mm in diameter — serves as the substrate throughout this process. Each cycle begins with a chemical pre-treatment, such as the thermal growth of a thin oxide film for electrical isolation. A photosensitive material called

resist is then deposited onto the wafer surface. An optical exposure system projects the pattern for the current layer — defined in a chromium film on a transparent quartz plate known as the *reticle*, or *photo-mask* — onto the resist. After development, the regions that received light are selectively dissolved (in the case of a positive resist), exposing the underlying material. This material then undergoes one of several process steps. *Plasma etching* selectively removes oxide or semiconductor material to define trenches and isolation regions. *Ion implantation* locally modifies the electrical properties of the silicon to form transistor junctions and doped wells. *Physical or chemical vapour deposition* lays down thin films of conductive metal — typically tungsten or copper — to form the interconnecting wiring that links the millions of functional elements stacked across all layers. After processing, the remaining resist is stripped and the cycle repeats for the next layer, until all layers are complete and the wafer is diced and packaged.

In the earliest days of the industry, the reticle was brought into direct physical contact with the resist-coated wafer during exposure — a technique known as *contact lithography*. While simple, this approach had two fundamental shortcomings. First, repeated physical contact progressively contaminated and damaged the delicate reticle pattern, necessitating its costly and time-consuming replacement. Second, a 1:1 projection offers no optical resolution advantage: any defect present on the reticle is transferred at full size onto the wafer. These limitations motivated a transition to *projection optical lithography*, in which a reduction lens images the reticle pattern onto the wafer from a safe working distance. The projection lens introduces a demagnification factor of four, meaning that features on the reticle are printed four times smaller on the wafer, immediately relaxing the demands on reticle fabrication. The minimum printable feature size, known as the *critical dimension* (CD), is governed by¹³:

$$\text{CD} = k \frac{\lambda}{\text{NA}}, \tag{1.0.1}$$

where λ is the illumination wavelength, NA is the numerical aperture of the projection lens, and k is a process-dependent factor. Reducing CD requires a shorter wavelength, a higher NA, and process improvements that lower k . This relation has driven a continuous push toward

shorter wavelengths — from mercury arc-lamp sources at 436 nm and 365 nm, through excimer lasers at 248 nm and 193 nm, to the extreme ultraviolet (EUV) sources at 13.5 nm now entering production.

Achieving a high numerical aperture, however, comes with a practical constraint. Since $NA = n \sin \vartheta$, where ϑ is the half-angle of the marginal ray accepted by the lens, attaining values of NA close to one in air requires the lens to subtend a very wide solid angle. Constructing a lens element that simultaneously covers the full area of a 300 mm wafer at high NA would require an optical element of impractical size and cost¹³. The industry's solution was to accept a smaller *exposure field* — typically 26×32 mm, corresponding to a single IC die — and expose the wafer in successive steps: after each die is exposed, the wafer stage repositions the wafer to the next die location and the exposure is repeated. This architecture defines the *wafer stepper*. In a stepper, each die is illuminated in a single, instantaneous full-field flash, with the stage brought to a complete stop before exposure commences.

While the stepper represented a decisive step forward, exposing an entire die simultaneously subjects the full field to any non-uniformities in the illumination intensity or lens aberrations. The *scanner* addresses this by replacing the full-field flash with a *scanning exposure*: the illumination system reveals only a narrow rectangular slit of the reticle, and the reticle stage sweeps the mask through this slit while the wafer stage moves synchronously in the opposite direction, maintaining the 4:1 demagnification relationship throughout^{3;13}. Because the reticle moves at four times the speed of the wafer, it traverses the full reticle pattern in the same time that the wafer moves one die height. This scanning motion offers significant advantages over full-field exposure: it averages out illumination dose non-uniformities across the exposure field, improves imaging uniformity at the field edges, and allows the use of a physically larger reticle that is easier to write with electron-beam pattern generators. After each die is scanned, the wafer is stepped to the next die position and the scan repeats, defining the *step-and-scan* cycle that gives modern lithographic tools their name. A throughput of over 175 wafers per hour —

corresponding to less than 0.2 s of exposure time per die — is not uncommon in state-of-the-art tools³.

A critical consequence of the multi-layer nature of IC fabrication is the requirement that each new layer be positioned with nanometre-level accuracy relative to all preceding layers — a specification known as *overlay*. The functional layers of a chip are interconnected by millions of tiny conductive pillars called *vias*; a misregistration between consecutive layers of even a few nanometres can misalign a via, break an intended connection, or create an unintended short circuit, rendering the IC defective¹³. Overlay requirements scale roughly in proportion with the CD and currently sit below 4 nm for leading-edge nodes. Overlay is therefore one of the primary technology drivers in scanner design, and it is governed directly by the positioning accuracy of the wafer and reticle stages during the scanning exposure.

The commercial performance of a scanner is ultimately measured in wafers per hour (WPH), since a machine whose capital cost can exceed \$150 million must generate revenue continuously. Although the exposure time per die is fixed by the optics and the process, the time spent stepping and *settling* — the interval between the end of a step motion and the moment the stage has damped its residual vibrations to within the nanometre-level positioning specification — represents a significant and controllable fraction of the total cycle time³, which places the allowable settle time below 10 ms. Every millisecond saved in settling directly translates into more dies exposed per hour and thus more chips delivered per unit time, with a lower cost per chip. Reducing settling time is therefore a central objective in scanner control engineering.

It is worth noting that, while EUV lithography at 13.5 nm has entered high-volume production for leading-edge nodes below 10 nm, deep ultraviolet (DUV) optical lithography at 193 nm remains the dominant technology for *mature nodes* — technology generations at 28 nm and above. Mature-node chips are the backbone of automotive electronics, industrial controllers, power management devices, analogue circuits, and the vast majority of consumer electronics.

The aggregate volume of wafers produced at mature nodes substantially exceeds that at leading-edge nodes, and is projected to remain so for the foreseeable future. Throughput improvements in DUV step-and-scan tools therefore continue to have a large global impact.

The key lever for reducing settling time — and thus for increasing scanner throughput — is the generation of *reference trajectories*: motion profiles that command the stage from its current position to the next scanning start-point with minimal excitation of mechanical resonances and the shortest possible residual settling tail. A well-designed trajectory limits the energy injected into the structural eigenmodes of the machine, reduces induced vibrations in the sensitive metrology and optical systems, and enables the stage to reach nanometre-level positioning accuracy well within the allocated window. The design of such trajectories is a problem of direct industrial relevance, and forms the central subject of this work.

1.1 Research Gap

Existing literature on trajectory design for wafer steppers evaluates candidate profiles exclusively through positioning metrics: tracking error, settling time, and the filtered exposure indices formalized in Chapter 3. What those metrics do not capture is the *morphology* of the failures into which these residual errors translate to on the wafer. Yield engineering, meanwhile, has long exploited that same morphology in the opposite direction: the pattern of the failing dies across the wafer — its *fingerprint* — is routinely used to diagnose which process step caused an observed defect¹⁰, and industrial datasets such as WM-811K¹⁶ catalogue hundreds of thousands of labelled examples of these patterns. No published methodology runs this relationship forward: from a given set of high-order kinematic bounds, predict the defect morphology they will produce on the wafer, and use that prediction as a cost function to optimize the trajectory. The absence of this link means that more aggressive acceleration profiles cannot be explored in an informed way, simultaneously limiting the achievable throughput and the wafer yield.

1.2 Hypothesis

The central hypothesis of this work is:

The kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ of a fourth-order reference trajectory can be optimized, subject to a throughput constraint, using as cost function the probability distribution over defect-morphology classes estimated by a convolutional neural network trained on industrial wafer fingerprints.

For the hypothesis to be testable, we decompose into three falsifiable sub-claims tied to a measurable criterion:

- H1 (Sensitivity)** The exposure-window synchronization error of the simulated machine responds monotonically and reproducibly to each active kinematic bound.
- H2 (Expressiveness)** The synthetic wafer defect maps produced from the simulation can be meaningfully separated by the CNN such that only the scanner-plausible classes activate.
- H3 (Optimality)** A population-based search over the four kinematic bounds, driven by the CNN cost, converges to profiles that overcome third-order and regular fourth-order baselines on the yield-throughput plane.

H1 is formulated because if sweeping a bound leaves exposure indices unchanged (or changes them erratically), it provides no gradient to effectively guide an optimizer. Similarly, **H2** is driven by the need to use the CNN probability distribution as a criterion to isolate the phenomenological causes behind the resulting error type on the wafer, while ensuring the simulation captures real scanner physics. Furthermore, **H3** establishes that a yield-optimal trajectory profile can be iteratively approximated to overcome conventional baseline limitations. Chapter 5 evaluates these three claims.

1.3 Objectives

The general objective is to develop a computational framework comprising a simulation of a lithography scanner and a convolutional neural network as a defect-evaluation model that can provide us with a yield-throughput plane over which a population-based optimizer can search for fourth-order trajectories that maximize estimated yield without penalizing throughput. This decomposes into six specific objectives:

- O1** Implement the multi-body dynamic model of² as a simulation restricted to the x - y plane, capable of following a parameterized trajectory profile.
- O2** Describe an analytical model to generate fourth-order trajectories from kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ and segment times.
- O3** Train and validate a convolutional neural network to classify the nine defect-morphology classes on WM-811K¹⁶.
- O4** Design a mapping from the dynamic error on a simulated traversal into a synthetic wafer defect map comparable with the CNN inputs.
- O5** Implement a population-based optimizer coupled to the simulation using the CNN’s probability distribution as a dynamic cost function to optimize the kinematic parameters of a reference trajectory.
- O6** Evaluate the framework through comparative scenarios quantifying the throughput-yield trade-off against third-order and regular fourth-order baselines.

Objectives O1–O5 are implemented and validated in this document (Chapters 2–5), including the exact fifteen-segment snap-limited generator together with its second- and third-order counterparts (Section 4.3). O6 is addressed by the optimization scenarios E3 and E4 and by the profile-order comparison of Experiment 7.

1.4 Outline of the Thesis

Chapter 2 develops the formulation of the multi-body model of the machine in state-space. Chapter 3 translates the industrial drivers into quantitative control objectives. Chapter 4 describes the simulation environment, the trajectory generator, the stage controller, the CNN pipeline. Chapter 5 reports the experiments performed to validate the simulated machine, the evaluation of the classifier and the trajectory search. The conclusion summarizes findings, limitations, and further comparative scenarios.

Chapter 2

Model

2.1 Introduction

As we established before, the accurate relative positioning between the reticle and the wafer is crucial for *step-and-scan* machines to produce functional ICs. To analyze these positioning requirements in a multi-body setting, we leverage existing literature² that proposes modeling the lithography scanner as a combination of three open-loop serial chains: the reticle chain ($j = 1$, blue), the optics chain ($j = 2$, purple), and the wafer chain ($j = 3$, red); the latter being the crux of the problem studied here.

The base frame serves as the common structural root for all three chains. In the reticle chain, a dual-stage architecture is employed to satisfy the competing requirements for travel range and precision. Initially, the long-stroke motor (LS) —of the planar moving-coil type¹⁴— consisting of four coil actuators arranged in a platform shape facilitates large-range travel (of at least 300 mm) with moderate micrometer accuracy in 6-DOF. However, being directly mounted to the base frame, it is inherently susceptible to vibrations propagated from the rest of the machine.

To achieve nanometric tracking accuracy, a short-stroke motor (SS) is appended to the long-stroke carrier. Since travel in this stage is limited up to 1 mm, it is realized with Lorentz actuators which provide contact-free, high-bandwidth positioning. Given that the Lorentz force depends entirely on the current within the actuator and is independent of the stator’s motion, it effectively decouples the end-effector from the base frame disturbances through magnetic suspension³. To minimize linearity errors and optimize force transfer, the long-stroke stage carrying the stator must maintain the optimal clearance to the magnets of the rest of the short-stroke actuator. This arrangement is replicated for the wafer chain.

The optics chain is attached to the base frame through a metrology frame which is in turn supported by pneumatic vibration isolators³. While the optics box includes adaptive optics with multiple elements, it is reduced in this framework to a single fixed lens. This allows the model to focus on machine compliance and structural vibration transfer rather than involving active optical control².

The physical connections between these stages, which constitute the primary transmission path for vibration, are represented by lumped-parameter spring-damper systems, according to the Kelvin-Voigt model. As its proponents state, this low-fidelity modeling choice is sufficient to study the dynamics between the multiple bodies during *step-and-scan* motion².

Figure 2.1 illustrates the structural topology and hierarchical arrangement within each chain. Horizontal Kelvin–Voigt elements $(K_{i,j}, C_{i,j})$ couple each stage laterally to the fixed structural reference depicted as a solid line in the color of the corresponding chain; in particular, elements (K_1, C_1) provide passive vibration isolation between $\mathcal{F}_0 \rightarrow \mathcal{F}_1$ through the solid green line that extends the base frame ($i = 1$) into the coupling to the first body ($i = 2$) in any j chain. The in-plane deviations $f_{i,j}$ represented by the horizontal blue double-headed arrows measure the relative displacement between each body. The dashed vertical centerline represents the nominal reference for $f_{i,j}^0$, when all relative coordinates are zero.

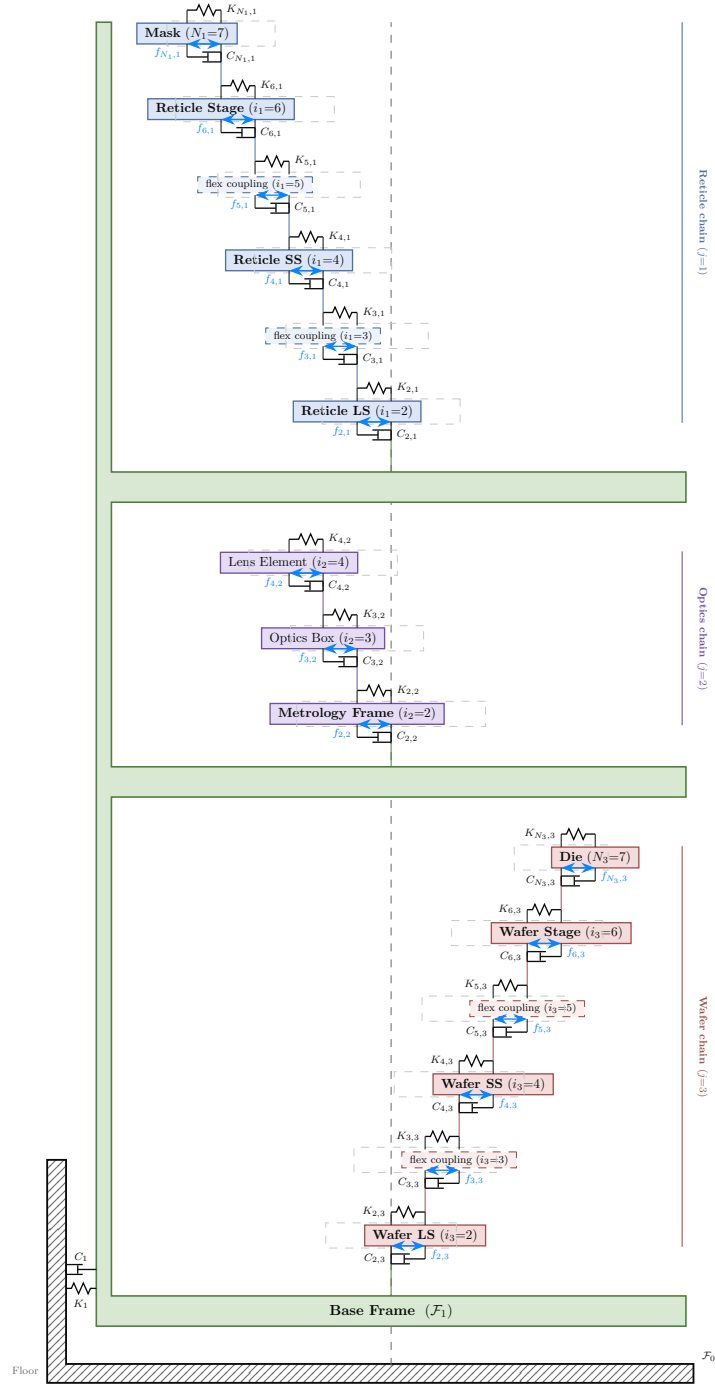


Figure 2.1: Lumped mass-spring-damper system of the three chains vertically stacked on the common base frame

2.2 Dynamics

Two distinct joint types arise: *actuated* stages ($i \in \{2, 4\}$), which are driven by servo motors and carry a near-infinite effective stiffness that keeps f_{i_j} locked to the nominal reference; and *flex* stages ($i \in \{3, 5, 6, 7\}$), which are passive compliant connections whose stiffness K_{i_j} and damping C_{i_j} are physical material properties of the mechanical coupling.

The full 6-DOF model is here restricted to in-plane motions^{1,4}, since rotations about the x and y axis, as well as translation on z are used mainly for focusing. The generalized coordinate used to describe the configuration of the i th stage within the j th chain is

$$q_{i_j} = \begin{bmatrix} f_{i_j} \\ \theta_{z_{i_j}} \end{bmatrix} \in \mathbb{R}^{3 \times 1} \quad (2.2.1)$$

where $f_{i_j} = [f_{x,i_j}, f_{y,i_j}]^T \in \mathbb{R}^2$ collects the in-plane translational displacements of stage i relative to its parent body in chain j , and $\theta_{z_{i_j}} \in \mathbb{R}$ is the corresponding rotation about the out-of-plane z -axis. The joints between bodies within a chain are modeled using a spring-damper system². The associated generalized mass-inertia matrix for each body is

$$M_{i_j} = \begin{bmatrix} m_{i_j} \mathbf{I}_2 & 0 \\ 0 & I_{z_{i_j}} \end{bmatrix} \in \mathbb{R}^{3 \times 3} \quad (2.2.2)$$

and the generalized stiffness and damping matrices $K_{i_j}, C_{i_j} \in \mathbb{R}^{3 \times 3}$ are block-diagonal, with a 2×2 translational block and a scalar rotational entry.

The absolute position $f_{i_j}^0$ of body i is expressed as the recursive sum of its relative displacement from its parent body along the chain from the floor frame $\mathcal{F}_0$. Thus, for bodies with $i \geq 2$ on any chain $j \in \{1, 2, 3\}$:

$$f_{i_j}^0 = f_{(i-1)_j}^0 + f_{i_j} \quad (2.2.3)$$

Since the rotation about the z -axis is a scalar, the angular position recurrence is simply additive:

$$\theta_{z_{i_j}}^0 = \theta_{z_{(i-1)_j}}^0 + \theta_{z_{i_j}} \quad (2.2.4)$$

From this, the translational velocity expression appears as:

$$\dot{f}_{i_j}^0 = \dot{f}_{(i-1)_j}^0 + \dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j} \quad (2.2.5)$$

Where $\omega_{(i-1)_j}$ is the angular velocity⁵ of $\mathcal{F}_{(i-1)_j}$ with respect to the floor, $\mathcal{F}_0$.

Taking the second derivative gives:

$$\begin{aligned} \ddot{f}_{i_j}^0 &= \ddot{f}_{(i-1)_j}^0 + \ddot{f}_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \dot{\omega}_{(i-1)_j} \times f_{i_j} + \omega_{(i-1)_j} \times [\dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j}] \\ &= \ddot{f}_{(i-1)_j}^0 + \ddot{f}_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \dot{\omega}_{(i-1)_j} \times f_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \omega_{(i-1)_j} \times [\omega_{(i-1)_j} \times f_{i_j}] \quad (2.2.6) \\ &= \ddot{f}_{(i-1)_j}^0 + \ddot{f}_{i_j} + 2 [\omega_{(i-1)_j} \times \dot{f}_{i_j}] + \dot{\omega}_{(i-1)_j} \times f_{i_j} + \omega_{(i-1)_j} \times [\omega_{(i-1)_j} \times f_{i_j}] \end{aligned}$$

Since we are concerned with planar movements in this work, we restrict the angular velocity of each body purely about the z -axis as $\omega = \dot{\theta}_z \hat{z}$, so that:

$$\omega_{(i-1)_j} = \dot{\theta}_{z(i-1)_j}^0 \hat{z}$$

We substitute the cross-products as matrix-vector products using the skew-symmetric matrix.

In the 6-DOF model², it appears as:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Clearly, here we can set $\omega_x = 0$ and $\omega_y = 0$, thus $\omega = [0, 0, \dot{\theta}_z]^T$ and the matrix becomes:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & 0 \\ \omega_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Considering our 2D displacement vector, we only take the upper-left 2×2 sub-matrix, and factoring out the scalar $\dot{\theta}_{z(i-1)_j}^0$:

$$[\omega_{(i-1)_j}] = \dot{\theta}_{z(i-1)_j}^0 \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (2.2.7)$$

such that $\omega_{(i-1)_j} \times f_{i_j} = [\omega_{(i-1)_j}] f_{i_j}$. Applying this matrix twice gives the centripetal term:

$$[\omega_{(i-1)_j}]^2 = - \left(\dot{\theta}_{z(i-1)_j}^0 \right)^2 \mathbf{I}_2 \quad (2.2.8)$$

so that $\omega_{(i-1)_j} \times [\omega_{(i-1)_j} \times f_{i_j}] = [\omega_{(i-1)_j}]^2 f_{i_j}$. Substituting and expanding (2.2.3) recursively from the floor frame, the translational kinematics become:

$$f_{i_j}^0 = f_1^0 + \sum_{k=2}^i f_{k_j} \quad (2.2.9)$$

$$\dot{f}_{i_j}^0 = \dot{f}_1^0 + \sum_{k=2}^i \dot{f}_{k_j} + \sum_{k=2}^i [\omega_{(k-1)_j}] f_{k_j} \quad (2.2.10)$$

$$\ddot{f}_{i_j}^0 = \ddot{f}_1^0 + \sum_{k=2}^i \ddot{f}_{k_j} + \sum_{k=2}^i \left[2 [\omega_{(k-1)_j}] \dot{f}_{k_j} + [\dot{\omega}_{(k-1)_j}] f_{k_j} + [\omega_{(k-1)_j}]^2 f_{k_j} \right] \quad (2.2.11)$$

Since the rotation $\theta_{z_{i_j}}$ is a scalar, no Coriolis or centripetal cross-terms arise in the angular kinematics. Expanding (2.2.4) recursively from the floor frame gives the rotational analogues:

$$\theta_{z_{i_j}}^0 = \theta_{z_1}^0 + \sum_{k=2}^i \theta_{z_{k_j}} \quad (2.2.12)$$

$$\dot{\theta}_{z_{i_j}}^0 = \dot{\theta}_{z_1}^0 + \sum_{k=2}^i \dot{\theta}_{z_{k_j}} \quad (2.2.13)$$

$$\ddot{\theta}_{z_{i_j}}^0 = \ddot{\theta}_{z_1}^0 + \sum_{k=2}^i \ddot{\theta}_{z_{k_j}} \quad (2.2.14)$$

Thus the equation of the angular motion of the i th body in the j th chain w.r.t $\mathcal{F}_0$ is:

$${}^0I_j \dot{\omega}_j + ([\omega_j]) {}^0I_j \omega_j = m_j^0 \quad (2.2.15)$$

Under the planar restriction $\omega_x = \omega_y = 0$ the gyroscopic term vanishes like:

$$([\omega_{i_j}] {}^0I_{i_j}) \omega_{i_j} = \begin{bmatrix} 0 & -\dot{\theta}_z & 0 \\ \dot{\theta}_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_z \end{bmatrix} = 0$$

Leaving only:

$$I_{z_{i_j}} \ddot{\theta}_{z_{i_j}}^0 = m_{z_{i_j}}^0 \quad (2.2.16)$$

where $I_{z_{i_j}}$ is the aggregated moment of inertia of all bodies from i to the end-effector N_j , referred to the z -axis of body i via Steiner's theorem, and $m_{z_{i_j}}^0 \in \mathbb{R}$ is the net applied moment acting on body i in chain j about the z -axis.

Steiner’s theorem and inertia aggregation. The parallel axis (Steiner’s) theorem states that the moment of inertia of a rigid body about any axis parallel to one through its center of mass (CoG) satisfies

$$I_z = \tilde{I}_z + m \|f\|^2, \quad (2.2.17)$$

where $\tilde{I}_z$ is the body’s own moment of inertia about the z -axis through its CoG, m is its mass, and $\|f\|^2 = f_x^2 + f_y^2$ is the squared in-plane distance between the two parallel axes. In a serial chain, the inertia of every downstream body must be shifted to the frame of the current body using (2.2.17). Following the characterization article² (Algorithm 1 therein), the aggregated inertia for the j th chain is computed recursively from the end-effector inward. In the planar restriction, the full 3×3 inertia matrix of the original algorithm reduces to a scalar z -axis inertia:

Algorithm 1 Planar inertia aggregation for the j th chain — planar reduction of Algorithm 1 in²

Require: CoG inertias $\tilde{I}_{z_{i_j}}$, masses m_{i_j} , relative positions f_{k_j} for $i = 2, \dots, N_j$

Ensure: Aggregated inertias ${}^0I_{z_{i_j}}$ for $i = 2, \dots, N_j$

```

1: Initialize:  ${}^0I_{z_{N_j}} \leftarrow \tilde{I}_{z_{N_j}}$ 
2: if  $2 \leq i < N_j$  then
3:   for  $k = N_j, \dots, i + 1$  do
4:      $\tilde{m}_{(k-1)_j} \leftarrow \sum_{b=k}^{N_j} m_{b_j}$  ▷ accumulated downstream mass
5:      ${}^0I_{z_{(k-1)_j}} \leftarrow {}^0I_{z_{k_j}} + \tilde{I}_{z_{(k-1)_j}} + \tilde{m}_{(k-1)_j} (f_{x,k_j}^2 + f_{y,k_j}^2)$  ▷ Steiner shift from frame  $k$  to
       frame  $k - 1$ 
6:   end for
7: end if

```

Here $\tilde{m}_{(k-1)_j} = \sum_{b=k}^{N_j} m_{b_j}$ is the total mass of all downstream bodies (from k to the end-effector), $\tilde{I}_{z_{(k-1)_j}}$ is the local inertia of body $k - 1$ about its own CoG, and $\tilde{m}_{(k-1)_j} (f_{x,k_j}^2 + f_{y,k_j}^2)$ is the Steiner correction that shifts the accumulated inertia from body k ’s CoG to body $(k - 1)$ ’s CoG. By the time the algorithm terminates, ${}^0I_{z_{2_j}}$ contains the inertia of the entire chain j

referred to the CoG of its first body.

The aggregated inertia of the base frame (body 1) applies the same shift one more time, from body 2's CoG to the base frame:

$${}^0I_{z_1} = \tilde{I}_{z_1} + \sum_{j=1}^3 \left({}^0I_{z_{2j}} + \tilde{m}_{1j} (f_{x,2j}^2 + f_{y,2j}^2) \right), \quad (2.2.18)$$

where $\tilde{m}_{1j} = \sum_{b=2}^{N_j} m_{b_j}$ is the total mass of chain j . This is the quantity denoted I_{z_0} in (2.2.22).

Expanding $\ddot{\theta}_{z_{i_j}}^0$ using (2.2.14):

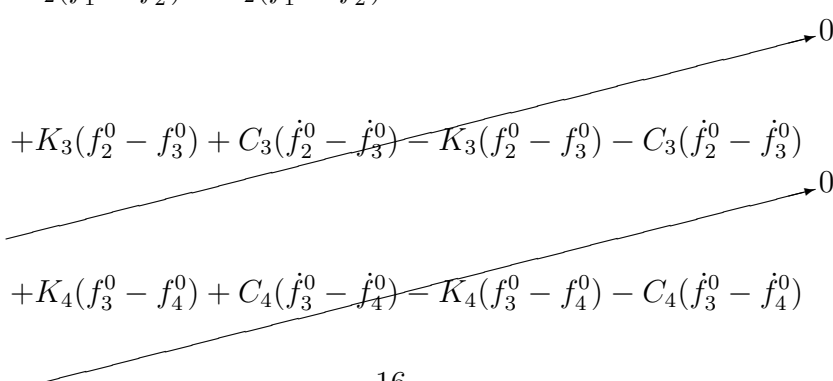
$$I_{z_{i_j}} \ddot{\theta}_{z_1}^0 + I_{z_{i_j}} \sum_{k=2}^i \ddot{\theta}_{z_{k_j}} = m_{z_{i_j}}^0 \quad (2.2.19)$$

2.2.1 For the base frame

To write the equations that govern the base frame, we start from Newton's equation $F = ma$. For every body ($i \geq 2$) within chain j , it connects to body $i - 1$ through a spring above and body $i + 1$ through the spring below:

$$\begin{aligned} m_{i_j} \ddot{f}_1^0 + m_{i_j} \sum_{k=2}^i \ddot{f}_{k_j} + m_{i_j} \sum_{k=2}^i \left[2 [\omega_{(k-1)_j}] \dot{f}_{k_j} + [\dot{\omega}_{(k-1)_j}] f_{k_j} + [\omega_{(k-1)_j}]^2 f_{k_j} \right] \\ - K_i (f_{(i-1)_j}^0 - f_{i_j}^0) - C_i (\dot{f}_{(i-1)_j}^0 - \dot{f}_{i_j}^0) + K_{i+1} (f_{i_j}^0 - f_{(i+1)_j}^0) + C_{i+1} (\dot{f}_{i_j}^0 - \dot{f}_{(i+1)_j}^0) = u_{i_j}^0 \end{aligned} \quad (2.2.20)$$

Summing (2.2.20) over all bodies in a chain (taking the optics chain $j = 2$, bodies $i = 2, 3, 4$ as a representative example), the internal spring forces cancel by Newton's third law:

$$\begin{aligned} & -K_2(f_1^0 - f_2^0) - C_2(\dot{f}_1^0 - \dot{f}_2^0) \\ & + K_3(f_2^0 - f_3^0) + C_3(\dot{f}_2^0 - \dot{f}_3^0) - K_3(f_2^0 - f_3^0) - C_3(\dot{f}_2^0 - \dot{f}_3^0) \\ & + K_4(f_3^0 - f_4^0) + C_4(\dot{f}_3^0 - \dot{f}_4^0) - K_4(f_3^0 - f_4^0) - C_4(\dot{f}_3^0 - \dot{f}_4^0) \end{aligned}$$


Only the boundary term $-K_2(f_1^0 - f_2^0) - C_2(\dot{f}_1^0 - \dot{f}_2^0)$ survives, which corresponds to the connection between body $i = 2$ and the base frame. This extends to any N_j -body chain. Recalling (2.2.3), $f_1^0 - f_{2_j}^0 = -f_{2_j}$, and following² with ${}^0K_{i_j} = {}^0R_{i_j} K_{i_j} {}^0R_{i_j}^T$ and ${}^0C_{i_j} = {}^0R_{i_j} C_{i_j} {}^0R_{i_j}^T$, the total translational equation of the base frame (absorbing the balance masses and collecting all chain contributions) is:

$$m_0 \ddot{f}_1^0 + {}^0\bar{C}_1 \dot{f}_1 + {}^0\bar{K}_1 f_1 - \sum_{j=1}^3 \left\{ {}^0\hat{C}_{2_j} \dot{f}_{2_j} + {}^0\hat{K}_{2_j} f_{2_j} \right\} = u_1^0 \quad (2.2.21)$$

where $m_0 = m_1 + \sum_{j=1}^3 \left(m_B^j + \sum_{i=2}^{N_j} m_{i_j} \right)$ is the total system mass, and the stiffness and damping matrices denote the 2×2 translational block. Following², ${}^0\bar{K}_1 = {}^0K_1$ and ${}^0\bar{C}_1 = {}^0C_1$, while the hatted chain couplings are ${}^0\hat{C}_{2_j} = {}^0C_{2_j}$ and ${}^0\hat{K}_{2_j} = {}^0K_{2_j} + {}^0C_{2_j}[\omega_1]$; the accents are defined in general in (2.2.29) below.

By the exact same cancellation argument applied to the moment balance about the z -axis — the moment exerted by the spring connecting body i to body $i - 1$ is $-K_{i_j} \theta_{z_{i_j}}$, so internal moments cancel identically — the rotational equation of the base frame is:

$$I_{z_0} \ddot{\theta}_{z_1}^0 + {}^0\bar{C}_1 \dot{\theta}_{z_1} + {}^0\bar{K}_1 \theta_{z_1} - \sum_{j=1}^3 \left\{ {}^0\hat{C}_{2_j} \dot{\theta}_{z_{2_j}} + {}^0\hat{K}_{2_j} \theta_{z_{2_j}} \right\} = \tau_1^0 \quad (2.2.22)$$

where $I_{z_0} = I_{z_1} + \sum_{j=1}^3 \left(I_{z_B}^j + \sum_{i=2}^{N_j} I_{z_{i_j}} \right)$ is the total moment of inertia of the system about the z -axis, ${}^0\bar{C}_{i_j}$ and ${}^0\bar{K}_{i_j}$ (and their hatted counterparts, which coincide with them for the scalar rotational entries) are the scalar rotational damping and stiffness of the joint connecting body i to its predecessor in chain j , and τ_1^0 is the net applied torque on the base frame.

2.2.2 For each chain

Recalling (2.2.20) and (2.2.19), for the first body in the chain ($i = 2$), which connects to the base frame above and to body $i = 3$ below, the translational equation is:

$$\bar{m}_{2_j} \ddot{f}_{2_j}^0 + {}^0\bar{C}_{2_j} \dot{f}_{2_j} + {}^0\bar{K}_{2_j} f_{2_j} - \left\{ {}^0\hat{C}_{3_j} \dot{f}_{3_j} + {}^0\hat{K}_{3_j} f_{3_j} \right\} = u_{2_j}^0 \quad (2.2.23)$$

and the rotational equation is:

$$I_{z_{2j}} \ddot{\theta}_{z_{2j}}^0 + {}^0\bar{C}_{2j} \dot{\theta}_{z_{2j}} + {}^0\bar{K}_{2j} \theta_{z_{2j}} - {}^0\hat{C}_{3j} \dot{\theta}_{z_{3j}} - {}^0\hat{K}_{3j} \theta_{z_{3j}} = \tau_{2j}^0 \quad (2.2.24)$$

For bodies $3 \leq i < N_j$, the same structure holds with bodies $2, \dots, i-1$ in between. The translational equation is:

$$\bar{m}_{i_j} \ddot{f}_{i_j}^0 + {}^0\bar{C}_{i_j} \dot{f}_{i_j} + {}^0\bar{K}_{i_j} f_{i_j} - \left\{ {}^0\hat{C}_{(i+1)_j} \dot{f}_{(i+1)_j} + {}^0\hat{K}_{(i+1)_j} f_{(i+1)_j} \right\} = u_{i_j}^0 \quad (2.2.25)$$

and the rotational equation is:

$$I_{z_{i_j}} \ddot{\theta}_{z_{i_j}}^0 + {}^0\bar{C}_{i_j} \dot{\theta}_{z_{i_j}} + {}^0\bar{K}_{i_j} \theta_{z_{i_j}} - {}^0\hat{C}_{(i+1)_j} \dot{\theta}_{z_{(i+1)_j}} - {}^0\hat{K}_{(i+1)_j} \theta_{z_{(i+1)_j}} = \tau_{i_j}^0 \quad (2.2.26)$$

Finally, for body $i = N_j$ (the end effector), there is no subsequent body in the chain. The translational and rotational equations are:

$$\bar{m}_{N_{j_j}} \ddot{f}_{N_{j_j}}^0 + {}^0\bar{C}_{N_{j_j}} \dot{f}_{N_{j_j}} + {}^0\bar{K}_{N_{j_j}} f_{N_{j_j}} = 0 \quad (2.2.27)$$

$$I_{z_{N_{j_j}}} \ddot{\theta}_{z_{N_{j_j}}}^0 + {}^0\bar{C}_{N_{j_j}} \dot{\theta}_{z_{N_{j_j}}} + {}^0\bar{K}_{N_{j_j}} \theta_{z_{N_{j_j}}} = 0 \quad (2.2.28)$$

2.2.3 Accented matrices and the nominal condition

Following Appendix A of², the accented matrices appearing in the equations above are not merely rotated joint matrices: they absorb the rotational cross-terms of (2.2.11) into effective damping and stiffness. With $[\omega]$ the planar skew form of (2.2.7) and $B_{i_j} = [\dot{\omega}_{i_j}] + [\omega_{i_j}]^2$:

$$\begin{aligned} \bar{m}_{i_j} &= \sum_{k=i}^{N_j} m_{k_j} \\ {}^0\bar{C}_{i_j} &= {}^0C_{i_j} + 2m_{i_j} [\omega_{(i-1)_j}] \\ {}^0\bar{K}_{i_j} &= {}^0K_{i_j} + {}^0C_{i_j} [\omega_{(i-1)_j}] + m_{i_j} B_{(i-1)_j} \\ {}^0\hat{C}_{(i+1)_j} &= {}^0C_{(i+1)_j} \\ {}^0\hat{K}_{(i+1)_j} &= {}^0K_{(i+1)_j} + {}^0C_{(i+1)_j} [\omega_{i_j}] \end{aligned} \quad (2.2.29)$$

The bar thus marks the matrices acting on the body’s *own* joint coordinates, which collect the Coriolis ($2m[\omega]$), damper-convection ($C[\omega]$), and Euler–centripetal (mB) contributions, while the hat marks the coupling to the *downstream* joint coordinates, which inherits only the convective damper term: the damper reacts to the inertial velocity $\dot{f} + [\omega]f$ of (2.2.10), so part of its force is proportional to position and migrates into the effective stiffness. The scalar rotational entries admit no such cross-terms in the planar restriction, so about the z -axis the accents are inert and $(\hat{\cdot}) = (\bar{\cdot}) = (\cdot)$. For $3 \leq i < N_j$ the model of² additionally couples body i to every upstream joint through the two-index matrices ${}^0\hat{C}_{i,j,k} = 2m_{i_j} [\omega_{(k-1)_j}]$ and ${}^0\hat{K}_{i,j,k} = m_{i_j} B_{(k-1)_j}$ for $2 \leq k < i$, omitted above.

Every one of these corrections is proportional to $\dot{\theta}_z$ or $\ddot{\theta}_z$ of a parent body. At the nominal operating condition $\dot{\theta}_z = \ddot{\theta}_z = 0$ — the regulated step-and-scan regime studied throughout this work — they vanish identically: the accented matrices reduce to the frame-rotated ${}^0K_{i_j}$ and ${}^0C_{i_j}$, the inertial accelerations of (2.2.11) reduce to sums of relative accelerations, and the equations of motion coincide with Eqs. (2.6)–(2.8) of² and become linear time-invariant. Away from the nominal condition, the corrections re-enter as the parameter-varying part $A_1(\Omega, \dot{\Omega})$ of the decomposition given in². Note that the scheduling variables Ω are themselves state variables — the $\dot{\theta}_z$ of each body — so the cross-terms are quadratic in the state (and the rotated matrices ${}^0K_{i_j}$, ${}^0C_{i_j}$ carry a further state dependence through ${}^0R_{i_j}(\theta_z)$): the machine model is, strictly, nonlinear, a *quasi*-LPV system whose frozen nominal dynamics are the LTI model above. The MuJoCo test bed used throughout this work assembles the full nonlinear multibody dynamics from this lumped topology directly, retaining these couplings without approximation; the LTI form above is recovered only as its frozen nominal dynamics at $\dot{\theta}_z = \ddot{\theta}_z = 0$, not as a separately integrated reduced model.

2.2.4 State-space formulation

Because each body possesses three planar degrees of freedom (f_x, f_y, θ_z) along with their corresponding time derivatives, it contributes exactly six states to the system. Following the machine description of² (Section 2.2 therein), the complete machine model comprises 6 bodies in the reticle chain ($N_1 = 7$), 3 bodies in the optics chain ($N_2 = 4$), and 6 bodies in the wafer chain ($N_3 = 7$). Including the common base frame (body 1), the system features a total of $1 + (N_1 - 1) + (N_2 - 1) + (N_3 - 1) = 16$ bodies. Consequently, the global state vector x encompasses $6 \times 16 = 96$ states:

$$x = \left[\underbrace{x_1, \dots, x_6}_{\text{base frame}}, \underbrace{x_7, \dots, x_{6N_1}}_{\text{chain 1: bodies } 2 \rightarrow N_1}, \underbrace{\quad}_{\text{chain 2}}, \underbrace{\quad}_{\text{chain 3}} \right]^T \quad (2.2.30)$$

The explicit mapping for each state is detailed in the table below. Here, p denotes the global sequential index of the i -th body in the j -th chain within the concatenated list of all bodies (starting with the base frame, followed sequentially by chains 1, 2, and 3). It is crucial to note that every positional entry within the state vector represents a *relative* joint displacement; the corresponding absolute inertial position $f_{i_j}^0$ can be reconstructed using the recursive summations from (2.2.9) to (2.2.12).

Solving (2.2.21) and (2.2.22) for the base-frame accelerations yields:

$$\ddot{f}_1^0 = -\frac{{}^0\bar{C}_1}{m_0} \dot{f}_1 - \frac{{}^0\bar{K}_1}{m_0} f_1 + \frac{1}{m_0} \sum_{j=1}^3 \left\{ {}^0\bar{C}_{2_j} \dot{f}_{2_j} + {}^0\bar{K}_{2_j} f_{2_j} \right\} + \frac{u_1^0}{m_0} \quad (2.2.31)$$

$$\ddot{\theta}_{z_1}^0 = -\frac{{}^0\bar{C}_1}{I_{z_0}} \dot{\theta}_{z_1} - \frac{{}^0\bar{K}_1}{I_{z_0}} \theta_{z_1} + \frac{1}{I_{z_0}} \sum_{j=1}^3 \left\{ {}^0\bar{C}_{2_j} \dot{\theta}_{z_{2_j}} + {}^0\bar{K}_{2_j} \theta_{z_{2_j}} \right\} + \frac{\tau_1^0}{I_{z_0}} \quad (2.2.32)$$

Table 2.2.1: State-space mapping for the global concatenated system vector

State-space Variable	Expression	Meaning
x_1	f_{x_1}	base frame x -position
x_2	f_{y_1}	base frame y -position
x_3	θ_{z_1}	base frame rotation
x_4	$\dot{f}_{x_1}$	base frame x -velocity
x_5	$\dot{f}_{y_1}$	base frame y -velocity
x_6	$\dot{\theta}_{z_1}$	base frame angular velocity
x_7	$f_{x_{2_1}}$	chain 1, body 2, x -position
x_8	$f_{y_{2_1}}$	chain 1, body 2, y -position
x_9	$\theta_{z_{2_1}}$	chain 1, body 2, rotation
x_{10}	$\dot{f}_{x_{2_1}}$	chain 1, body 2, x -velocity
x_{11}	$\dot{f}_{y_{2_1}}$	chain 1, body 2, y -velocity
x_{12}	$\dot{\theta}_{z_{2_1}}$	chain 1, body 2, angular velocity
$\vdots$	$\vdots$	$\vdots$
x_{6p-5}	$f_{x_{i_j}}$	chain j , body i , x -position
x_{6p-4}	$f_{y_{i_j}}$	chain j , body i , y -position
x_{6p-3}	$\theta_{z_{i_j}}$	chain j , body i , rotation
x_{6p-2}	$\dot{f}_{x_{i_j}}$	chain j , body i , x -velocity
x_{6p-1}	$\dot{f}_{y_{i_j}}$	chain j , body i , y -velocity
x_{6p}	$\dot{\theta}_{z_{i_j}}$	chain j , body i , angular velocity
$\vdots$	$\vdots$	$\vdots$

Expressed in the standard matrix form $\dot{x} = Ax + Bu$, the first six rows governing the base-frame dynamics are:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \dots \\ -\frac{K_{1,xx}}{m_0} & -\frac{K_{1,xy}}{m_0} & 0 & -\frac{C_{1,xx}}{m_0} & -\frac{C_{1,xy}}{m_0} & 0 & \frac{K_{21,xx}}{m_0} & \frac{K_{21,xy}}{m_0} & 0 & \frac{C_{21,xx}}{m_0} & \dots \\ -\frac{K_{1,yx}}{m_0} & -\frac{K_{1,yy}}{m_0} & 0 & -\frac{C_{1,yx}}{m_0} & -\frac{C_{1,yy}}{m_0} & 0 & \frac{K_{21,yx}}{m_0} & \frac{K_{21,yy}}{m_0} & 0 & \frac{C_{21,yx}}{m_0} & \dots \\ 0 & 0 & -\frac{K_{1,zz}}{I_{z_0}} & 0 & 0 & -\frac{C_{1,zz}}{I_{z_0}} & 0 & 0 & \frac{K_{21,zz}}{I_{z_0}} & 0 & \dots \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_0} \\ 0 \\ 0 \end{bmatrix} \quad (2.2.33)$$

General State-Space Form for the Chain Bodies

To systematically construct the global state-space matrices, we must reframe the equations of motion for the chain bodies in terms of the state vector variables. We map the physical planar coordinates of the p -th sequential body to the local state vectors: translational position $\mathbf{x}_{2p-1} = [x_{6p-5}, x_{6p-4}]^T$, translational velocity $\dot{\mathbf{x}}_{2p-1} = [\dot{x}_{6p-5}, \dot{x}_{6p-4}]^T$, rotational position $\mathbf{x}_{2p} = x_{6p-3}$, and rotational velocity $\dot{\mathbf{x}}_{2p} = \dot{x}_{6p-3}$.

The state derivatives are dictated by the relative accelerations between adjacent bodies. By applying the kinematic identity $\ddot{f}_{i_j} = \ddot{f}_{i_j}^0 - \ddot{f}_{(i-1)_j}^0$ from (2.2.11), we determine the interactive forces acting on each body within its parent's reference frame. Let p represent the global index of body i in chain j , while $p-1$ and $p+1$ denote the indices of its immediate upstream and downstream neighbors respectively (with the base frame positioned at $p=1$).

For the **first body in each chain** ($i=2$, index p , upstream index 1), the translational state

derivatives ($\ddot{\mathbf{x}}_{2p-1} = \ddot{f}_{2j}$) and rotational state derivatives ($\ddot{\mathbf{x}}_{2p} = \ddot{\theta}_{z_{2j}}$) take the form:

$$\begin{aligned}\ddot{\mathbf{x}}_{2p-1} = & \frac{1}{\bar{m}_{2j}} \left[u_{2j}^0 - {}^0\bar{C}_{2j} \dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{2j} \mathbf{x}_{2p-1} + {}^0\hat{C}_{3j} \dot{\mathbf{x}}_{2p+1} + {}^0\hat{K}_{3j} \mathbf{x}_{2p+1} \right] \\ & - \frac{1}{m_0} \left[u_1^0 - {}^0\bar{C}_1 \dot{\mathbf{x}}_1 - {}^0\bar{K}_1 \mathbf{x}_1 + \sum_{k=1}^3 \left({}^0\hat{C}_{2k} \dot{\mathbf{x}}_{2p_{2k}-1} + {}^0\hat{K}_{2k} \mathbf{x}_{2p_{2k}-1} \right) \right]\end{aligned}\quad (2.2.34)$$

$$\begin{aligned}\ddot{\mathbf{x}}_{2p} = & \frac{1}{I_{z_{2j}}} \left[\tau_{2j}^0 - {}^0\bar{c}_{\theta,2j} \dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,2j} \mathbf{x}_{2p} + {}^0\hat{c}_{\theta,3j} \dot{\mathbf{x}}_{2p+2} + {}^0\hat{k}_{\theta,3j} \mathbf{x}_{2p+2} \right] \\ & - \frac{1}{I_{z_0}} \left[\tau_1^0 - {}^0\bar{C}_1 \dot{\mathbf{x}}_2 - {}^0\bar{K}_1 \mathbf{x}_2 + \sum_{k=1}^3 \left({}^0\hat{c}_{\theta,2k} \dot{\mathbf{x}}_{2p_{2k}} + {}^0\hat{k}_{\theta,2k} \mathbf{x}_{2p_{2k}} \right) \right]\end{aligned}\quad (2.2.35)$$

For a **general intermediate body** ($2 < i < N_j$, index p), the dynamics are shaped by upstream, local, and downstream interactions:

$$\begin{aligned}\ddot{\mathbf{x}}_{2p-1} = & \frac{1}{\bar{m}_{i_j}} \left[u_{i_j}^0 - {}^0\bar{C}_{i_j} \dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{i_j} \mathbf{x}_{2p-1} + {}^0\hat{C}_{(i+1)_j} \dot{\mathbf{x}}_{2p+1} + {}^0\hat{K}_{(i+1)_j} \mathbf{x}_{2p+1} \right] \\ & - \frac{1}{\bar{m}_{(i-1)_j}} \left[u_{(i-1)_j}^0 - {}^0\bar{C}_{(i-1)_j} \dot{\mathbf{x}}_{2p-3} - {}^0\bar{K}_{(i-1)_j} \mathbf{x}_{2p-3} + {}^0\hat{C}_{i_j} \dot{\mathbf{x}}_{2p-1} + {}^0\hat{K}_{i_j} \mathbf{x}_{2p-1} \right]\end{aligned}\quad (2.2.36)$$

$$\begin{aligned}\ddot{\mathbf{x}}_{2p} = & \frac{1}{I_{z_{i_j}}} \left[\tau_{i_j}^0 - {}^0\bar{c}_{\theta,i_j} \dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,i_j} \mathbf{x}_{2p} + {}^0\hat{c}_{\theta,(i+1)_j} \dot{\mathbf{x}}_{2p+2} + {}^0\hat{k}_{\theta,(i+1)_j} \mathbf{x}_{2p+2} \right] \\ & - \frac{1}{I_{z_{(i-1)_j}}} \left[\tau_{(i-1)_j}^0 - {}^0\bar{c}_{\theta,(i-1)_j} \dot{\mathbf{x}}_{2p-2} - {}^0\bar{k}_{\theta,(i-1)_j} \mathbf{x}_{2p-2} + {}^0\hat{c}_{\theta,i_j} \dot{\mathbf{x}}_{2p} + {}^0\hat{k}_{\theta,i_j} \mathbf{x}_{2p} \right]\end{aligned}\quad (2.2.37)$$

Finally, for the **end-effector body** ($i = N_j$, index p), the downstream force terms naturally vanish as there are no subsequent bodies:

$$\begin{aligned}\ddot{\mathbf{x}}_{2p-1} = & \frac{1}{\bar{m}_{N_{j_j}}} \left[u_{N_{j_j}}^0 - {}^0\bar{C}_{N_{j_j}} \dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{N_{j_j}} \mathbf{x}_{2p-1} \right] \\ & - \frac{1}{\bar{m}_{(N_j-1)_j}} \left[u_{(N_j-1)_j}^0 - {}^0\bar{C}_{(N_j-1)_j} \dot{\mathbf{x}}_{2p-3} - {}^0\bar{K}_{(N_j-1)_j} \mathbf{x}_{2p-3} + {}^0\hat{C}_{N_{j_j}} \dot{\mathbf{x}}_{2p-1} + {}^0\hat{K}_{N_{j_j}} \mathbf{x}_{2p-1} \right]\end{aligned}\quad (2.2.38)$$

$$\begin{aligned}\ddot{\mathbf{x}}_{2p} = & \frac{1}{I_{z_{N_{j_j}}}} \left[\tau_{N_{j_j}}^0 - {}^0\bar{c}_{\theta,N_{j_j}} \dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,N_{j_j}} \mathbf{x}_{2p} \right] \\ & - \frac{1}{I_{z_{(N_j-1)_j}}} \left[\tau_{(N_j-1)_j}^0 - {}^0\bar{c}_{\theta,(N_j-1)_j} \dot{\mathbf{x}}_{2p-2} - {}^0\bar{k}_{\theta,(N_j-1)_j} \mathbf{x}_{2p-2} + {}^0\hat{c}_{\theta,N_{j_j}} \dot{\mathbf{x}}_{2p} + {}^0\hat{k}_{\theta,N_{j_j}} \mathbf{x}_{2p} \right]\end{aligned}\quad (2.2.39)$$

Chapter 3

Control Objectives

3.1 Introduction

In Chapter 1 we identified overlay and throughput as the two primary drivers of scanner design, followed in Chapter 2 by the modeling of its multi-body dynamics. This chapter focuses on mapping those industrial requirements into quantitative objectives that any candidate reference trajectory must satisfy. Here we explore in detail how the synchronization error arises between the wafer and the reticle stages, examine the filtered metrics through which the industry evaluates it, and establish the formal statement of the optimization problem.

3.2 Synchronization Error

During the scanning exposure, the projection optics image the reticle onto the wafer with a demagnification factor of $\alpha = 0.25$. Thus, the image of the reticle pattern is stationary on the wafer surface only if the wafer follows the reticle motion scaled by α at every instant. The degradation of the printed pattern is therefore not due to the individual tracking error of either stage, but their scaled difference. Let $r_w(t)$ and $r_r(t)$ denote the reference positions of the wafer

and reticle stages respectively along the scanning direction, and $y_w(t)$, $y_r(t)$ their measured positions. The *synchronization error* is defined as³

$$e_{syn}(t) = (y_w(t) - r_w(t)) - \alpha(y_r(t) - r_r(t)). \quad (3.2.1)$$

Naturally, a common-mode mode disturbance that displaces both stages consistently with the projection ratio leaves e_{syn} unaffected. Conversely, any differential motion between the chains — exacerbated by the lightly damped structural resonance — is transferred directly to the printed image.

3.3 Exposure Metrics: MA and MSD

For an area of photoresist to become exposed, the illumination dose must overcome a threshold. Despite direct exposure during the scanning motion, the photoresist on the wafer does not respond to the instantaneous value of e_{syn} but to its aggregate effect over the time a given point stays below the light source. Therefore, industry practice evaluates the synchronization error through two moving-window filters matched to the exposure time t_{exp} of one point on the wafer³:

- First, the **moving average**

$$\text{MA}(t_k) = \frac{1}{N} \sum_{n=0}^{N-1} e_{syn}(t_{k-n}) \quad (3.3.1)$$

measures the mean displacement of the image and manifests as *overlay* error or the misalignment of the printed pattern with respect to the previous layer.

- Second, the **moving standard deviation**

$$\text{MSD}(t_k) = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} \left(e_{syn}(t_{k-n}) - \text{MA}(t_k) \right)^2} \quad (3.3.2)$$

measures the smearing of the aerial image around that mean position and manifests as *fading*, a loss of contrast that distorts the printed feature.

For the 38 nm half-pitch process, the specifications adopted² are

$$\text{MA} \in [-1.25, +2] \text{ nm}, \quad \text{MSD} \leq 7 \text{ nm}, \quad (3.3.3)$$

evaluated at every instant during the exposure window³. These bounds must be met while the stage sustains the required scanning velocity, operating within a settling time budget of under 10 ms to preserve wafer throughput. On the two-stage machine modelled here they are not a fixed pass/fail but the tight end of a trade curve: Chapter 5 shows that a sufficiently smooth fourth-order trajectory meets them — both MSD and MA, to the nanometre — at the cost of a longer scan, so the specification and the throughput budget are the two ends of the optimization this thesis performs.

3.4 Problem Statement

The reference trajectories studied here are fourth-order profiles parameterized by the kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ on velocity, acceleration, jerk, and snap respectively. These four variables set both the duration of the motion — and hence throughput — and the frequency content of the excitation injected into the machine structure, and thereby the synchronization error during exposure.

Conventional trajectory design evaluates candidates using positioning metrics like settling time, peak error, or the MA/MSD bounds of (3.3.3). Such metrics make it possible to predict which dies will be defective, but reveal nothing about the *morphology* of the failures that accumulate across the wafer. Wafer fabs routinely exploit the converse relationship to diagnose which process step caused a particular spatial pattern of failing dies (the *fingerprint*)¹⁰. However, the forward direction remains an open problem as no published methodology predicts, from a given set of trajectory parameters, the defect pattern they will produce, nor uses that prediction as the cost function during trajectory optimization. This thesis addresses precisely that forward direction.

To evaluate the impact of a candidate trajectory on the overall process yield, the resulting wafer defect map is classified by a convolutional neural network trained on the WM811K dataset¹⁶. The network estimates a probability P_c for each defect morphology class c , and the trajectory cost is defined as the weighted sum of these probabilities:

$$\mathcal{J} = \sum_{c \in \mathcal{C}} w_c P_c, \quad (3.4.1)$$

where w_c is the economic weight of class c ^{8;9;7} and $\mathcal{C}$ is restricted to the classes plausibly caused by scanner motion (namely *none*, *Scratch*, *Edge-Loc*, and *Random*). The remaining classes (*Center*, *Donut*, *Edge-Ring*, *Loc*, *Near-full*) originate in subsequent steps of the manufacturing process. If they were to be activated, it would indicate a modeling error rather than a trajectory effect; therefore, this restriction is used as a validation criterion for the trajectory-to-defect mapping in Chapter 5.

The optimization problem is then formulated as

$$\min_{(v_{\max}, a_{\max}, j_{\max}, s_{\max})} \mathcal{J} \quad \text{subject to} \quad T_{wafer} \leq T_{budget}, \quad (3.4.2)$$

where T_{wafer} is the total time to traverse a wafer determined by the trajectory parameters, and T_{budget} is the cycle-time budget corresponding to the target throughput. The central hypothesis of this work is that (3.4.2) can be solved using a data-driven cost function of the form (3.4.1), producing trajectory parameters that satisfy throughput constraints while minimizing scanner-induced defect costs.

Chapter 4

Methodology

4.1 Introduction

Trajectory generation is a well-studied problem. Trajectories with smooth higher-order derivatives are known to reduce the excitation of poorly damped structural modes. To evaluate the yield impact of our proposed trajectories without requiring physical scanner hardware, we need a simulation testbed where they can be executed and analyzed.

4.2 Simulation

In this section we describe the implementation of the dynamics model² described in Chapter 2 using the MuJoCo¹⁵ physics engine for Python. The simulation serves as a testbed for validating the proposed trajectories and evaluating the synthetic wafer defect maps generated by their traversal.

The MuJoCo¹⁵ Python bindings allow us to define the context of our simulation from an XML string by calling the `mujoco.MjModel.from_xml_string(MODEL_XML)` method. We extend from MuJoCo's `<worldbody>` tag on which we place a root `<body>` tag, representing our *base*

frame; this allows us to organize the machine in the simulation using the three chain hierarchical tree. Table 4.2.1 maps the specific names used in the MuJoCo XML to their counterparts in the structural diagram of Figure 2.1.

Table 4.2.1: Mapping of MuJoCo XML bodies to the model components

<body> name	property	Model body
base_frame		Base Frame ($\mathcal{F}_1$)
w_ls_act		Wafer LS ($i_3 = 2$)
w_ls_flex		flex coupling ($i_3 = 3$)
w_ss_act		Wafer SS ($i_3 = 4$)
w_ss_flex		flex coupling ($i_3 = 5$)
w_stage		Wafer Stage, passive ($i_3 = 6$)
wafer		Die ($N_3 = 7$)
r_ls_act		Reticle LS ($i_1 = 2$)
r_ls_flex		flex coupling ($i_1 = 3$)
r_ss_act		Reticle SS ($i_1 = 4$)
r_ss_flex		flex coupling ($i_1 = 5$)
r_stage		Reticle Stage, passive ($i_1 = 6$)
mask		Mask ($N_1 = 7$)
metro_frame		Metrology Frame ($i_2 = 2$)
optics_box		Optics Box ($i_2 = 3$)
lens		Lens Element ($i_2 = 4$)

The simulation environment is defined through several specialized XML attributes that handle different aspects of the physical world. Table 4.2.2 summarizes the function of each tag within the context of our lithography scanner model.

The mechanical properties of the couplings, represented by the Kelvin-Voigt elements, are

Table 4.2.2: MuJoCo XML tag functions in the scanner simulation

Tag	Function
<code><worldbody></code>	Defines the inertial root, the floor plane ($\mathcal{F}_0$).
<code><body></code>	Creates a rigid body with its own local coordinate system.
<code><inertial></code>	Assigns mass m_{i_j} and the aggregated inertia tensor I_{i_j} .
<code><joint></code>	Implements the degrees of freedom f_{i_j} and $\theta_{z_{i_j}}$.
<code><geom></code>	Defines the material (granite, silicon, glass) and collision shapes.
<code><actuator></code>	Models the Lorentz and moving-coil motors via servo controllers.

implemented via the `stiffness` and `damping` attributes of the `<joint>` tags. To begin, the common base frame ($\mathcal{F}_1$) is anchored to the world frame ($\mathcal{F}_0$) via a 3-DOF joint (two prismatic, one revolute) representing the isolation system with stiffness K_1 and damping C_1 , as seen in Eq. (2.2.21) and (2.2.22). For the two actuated stages of each chain (long-stroke and short-stroke), where the physical stiffness is replaced by the controller’s active response, the simulation uses high K_P values ($\mathbf{k}_p = 10^7 \text{ N/m}$, to approximate the stiffness of a rigid servo motor while maintaining numerical stability with the implicit integrator at a timestep of $dt = 10^{-4} \text{ s}$).

Tables 4.2.3, 4.2.4, and 4.2.5 illustrate the distribution of these variables for the three chains.

To successfully adapt our planar reduction of the reference model² into MuJoCo’s multibody solver, two numerical adaptations are in order:

1. Native joint damping against absolute velocity would exert artificial drag ($K_V/K_P = 10 \text{ mm per m/s}$) during cruise, so we apply the velocity gain K_V to the tracking-error velocity ($\dot{r} - \dot{y}$) within the control loop to represent the frictionless air bearings.
2. Actuators transmit reaction forces to the parent body by default, so we apply reaction cancellation to the base frame at each integration step to represent balance masses used to absorb long-stroke reaction forces³, which the analytical model² represents as external

Table 4.2.3: Wafer chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Floor $\rightarrow$ Base Frame	K1	C1
Base $\rightarrow$ LS Stage	KP_ACT	KV_ACT_TRANS
LS Stage $\rightarrow$ LS Flex	K_LS_F	C_LS_F
LS Flex $\rightarrow$ SS Stage	KP_ACT	KV_ACT_TRANS
SS Stage $\rightarrow$ SS Flex	K_SS_F	C_SS_F
SS Flex $\rightarrow$ Stage	K_ST_F	C_ST_F
Stage $\rightarrow$ Die	K8_W	C8_W

Table 4.2.4: Reticle chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Floor $\rightarrow$ Base Frame	K1	C1
Base $\rightarrow$ LS Stage	KP_ACT	KV_ACT_TRANS
LS Stage $\rightarrow$ LS Flex	K_LS_F	C_LS_F
LS Flex $\rightarrow$ SS Stage	KP_ACT	KV_ACT_TRANS
SS Stage $\rightarrow$ SS Flex	K_SS_F	C_SS_F
SS Flex $\rightarrow$ Stage	K_ST_F	C_ST_F
Stage $\rightarrow$ Mask	K8_R	C8_R

Table 4.2.5: Optics chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Base Frame $\rightarrow$ Metro Frame	K2_0	C2_0
Metro Frame $\rightarrow$ Optics Box	K3_0	C3_0
Optics Box $\rightarrow$ Lens Element	K4_0	C4_0

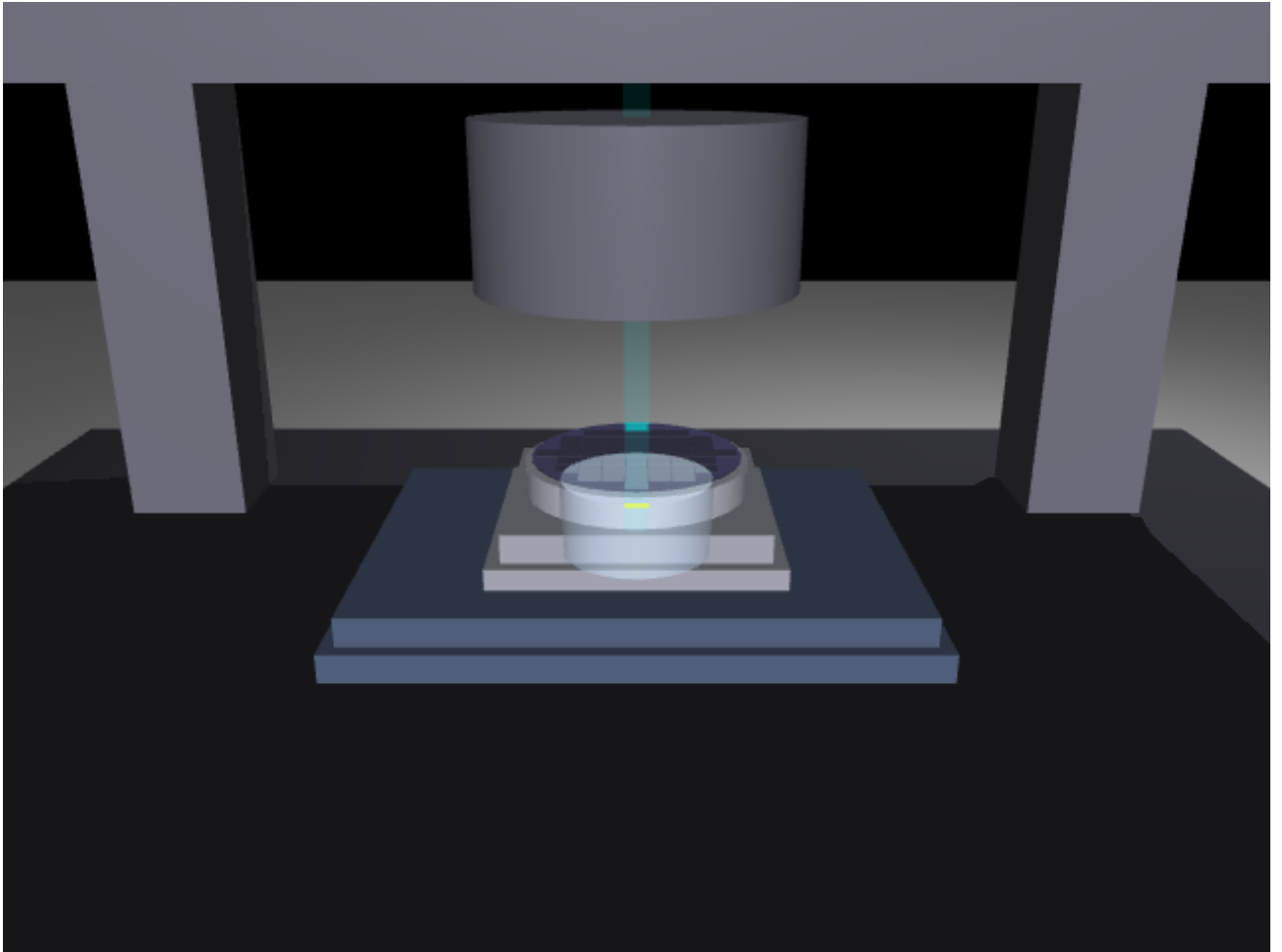


Figure 4.1: Screenshot of the MuJoCo simulation in the midst of the scanning motion, showing the wafer and the optics chain under the metrology frame

stage forces.

Synchronized scanning motion is commanded with nominal cruise velocities of $v_{w,\text{scan}} = 0.8 \text{ m/s}$ and $v_{r,\text{scan}} = 3.2 \text{ m/s}$. Under acceleration transients up to $a_{w,\text{max}} = 20 \text{ m/s}^2$, this setup captures the full-state dynamic interactions derived in Section 2.2.

Baseline control configurations (such as Case 1 in²) are realized on this plant by disabling active fine-stage feedback loops.

4.3 Reference Trajectory Generation

The trajectories commanded to the stages are **fourth-order profiles**: piecewise-polynomial motions in which position, velocity, acceleration, and jerk are all continuous, and the fourth derivative — the *snap* — switches between $\pm s_{\text{max}}$ and zero¹¹. Within each segment the position evolves as

$$p(t) = p_0 + v_0 t + \frac{a_0}{2} t^2 + \frac{j_0}{6} t^3 + \frac{s}{24} t^4, \quad (4.3.1)$$

where (p_0, v_0, a_0, j_0) are the boundary conditions inherited from the previous segment and $s \in \{-s_{\text{max}}, 0, +s_{\text{max}}\}$. A symmetric point-to-point motion consists of fifteen such segments, arranged so that none of the kinematic bounds $(v_{\text{max}}, a_{\text{max}}, j_{\text{max}}, s_{\text{max}})$ is exceeded.

We implement an analytical trajectory generator that structures the motion profile across fifteen segments (seven per acceleration ramp) where snap is held constant. Physical limitations are addressed by the following considerations from¹¹:

- When the jerk plateau cannot be reached, the effective jerk reduces to $\sqrt{a_{\text{max}} s_{\text{max}}}$
- If the acceleration limit is unreachable, then the effective acceleration follows from the corresponding quadratic equation; if the jerk phase collapses as well, the snap-phase duration instead follows from a cube-root solve

- If the total travel distance is too short to accommodate the required transition ramps, then the cruise velocity is iteratively reduced via bisection.

To streamline the trials, this same algorithmic structure is used to derive the lower order profiles: i.e., a second-order, acceleration-bounded trapezoidal profile and a third-order, seven-segment jerk-bounded S-curve.

The order of the motion profile determines the high-frequency content of the force commanded to initiate such motion: a second-order profile demands instantaneous jumps in acceleration, while a third-order profile is discontinuous in jerk⁴, so both inject energy across a broad frequency band. A fourth-order trajectory, whose snap is bounded as noted above, is reported to roll off the excitation spectrum at -80 dB per decade¹¹ — a theoretical asymptotic for constant-snap profiles, not separately verified spectrally in this work — reducing the response of poorly damped structural modes such as those of the end-effectors, the reticle and the wafer. The settling period after each step^{11:1} is thereby minimized, which is essential for the constant-velocity scanning phase, where ideally no force beyond the viscous terms is present.

4.4 Stage Feedback Control

The stage controllers are evaluated across four incremental configurations, following². Each setup introduces a specific feedforward or feedback component to isolate its impact on exposure performance in Chapter 5.

In Case 1, each long-stroke stage is regulated in closed loop by a collocated PD controller acting on the long-stroke encoder $y_c(t)$:

$$u(t) = K_P e(t) + K_V \dot{e}(t), \quad (4.4.1)$$

where $e(t) = r(t) - y_c(t)$ is the long-stroke tracking error, with gains $K_P = 10^7$ N/m and

$K_V = 10^5 \text{ N}\cdot\text{s}/\text{m}$. Position and velocity are read from the long-stroke stage rather than the end-effector because non-collocated control through the lightly damped reticle coupling ($\zeta \approx 0.04$) destabilizes the loop. Consequently, the lag between the long-stroke stage and the end-effector remains uncompensated and appears directly in the synchronization error. On this frictionless plant, this lag is dominated by its acceleration-proportional component: servo deflection $m a/K_P$ and quasi-static coupling stretch $m_d a/K$ carrying the downstream mass m_d . This configuration serves as the baseline validated in Experiment 1 (Section 5.2).

Since both lag components are functions of the known reference kinematics, in Case 2 they are cancelled by commanding the long stroke *ahead* of the reference:

$$r_{LS}(t) = r(t) - (\ell_v \dot{r}(t) + \ell_a \ddot{r}(t)), \quad (4.4.2)$$

so that the end-effector may land on the desired position. The coefficients (ℓ_v, ℓ_a) for each chain are identified by performing iterative least-squares regressions on the end-effector lag with respect to reference velocity and acceleration. Each resulting fit is added to the feedforward control loop while subsequent passes repeat the regression on the remaining residuals. Three iterations are required to prevent post-acceleration coupling vibrations from skewing the quasi-static fit. This is a model-free equivalent to the spring-compensation configuration of the reference study².

The error remaining after feedforward is a non-repeatable transient response that can only be suppressed through feedback. Thus, in Case 3, the short-stroke stage of each chain implements a closed position loop on the end-effector error $e_f(t)$, commanding the short-stroke actuator directly. In actual step-and-scan systems, this measurement is readily available from interferometry, as required for exposure metrology. Case 3a uses a linear PI law with anti-windup, whereas Case 3b replaces the linear integrator with a Hybrid Integrator–Gain System (HIGS)⁶, given by:

$$\dot{x}_1 = \omega_h e_f \quad \text{while } e_f x_1 \geq x_1^2/k_h, \quad x_1 = k_h e_f \quad \text{otherwise}, \quad (4.4.3)$$

with output $u_f = k_p e_f + \omega_i \int x_1 dt$. By resetting its state to the gain line $k_h e_f$ whenever the sector condition is violated, the HIGS element provides integral action with roughly 38° less phase lag

than a linear integrator, relaxing the overshoot-versus-stiffness limitation of linear designs — the property that motivates nonlinear integrators for wafer stages. Both fine loops saturate at the physical travel of the short stroke, with conditional integration preventing windup during large transients. The loop gains are tuned empirically over a grid up to the stability boundary of the simulated plant: the PI loop uses $k_p = 16$, $k_i = 480 \text{ s}^{-1}$, and the HIGS loop $k_p = 16$, $\omega_i = 2\pi \cdot 200 \text{ rad/s}$, $\omega_h = 2\pi \cdot 80 \text{ rad/s}$, $k_h = 1$; one step further in gain destabilizes both loops against the 180 Hz flexible-chain modes. The HIGS element sustains an integrator frequency roughly 42× above the PI’s equivalent k_i/k_p before reaching that boundary — the phase advantage of the nonlinear integrator expressed as usable gain. The fine-stage authority is ultimately bounded by the short-stroke actuator’s own velocity loop (pole at $K_P/K_V = 100 \text{ rad/s}$), a structural property of this generic replication that Chapter 5 quantifies.

4.5 Population-Based Trajectory Search

The optimization problem (3.4.2) is searched with particle swarm optimization (PSO) over the four kinematic bounds. Each particle holds a position $\mathbf{x} = (v_{\max}, a_{\max}, j_{\max}, s_{\max})$ and is updated by the standard rule

$$\mathbf{u}_{k+1} = \omega \mathbf{u}_k + c_1 r_1 (\mathbf{p}_k - \mathbf{x}_k) + c_2 r_2 (\mathbf{g}_k - \mathbf{x}_k), \quad \mathbf{x}_{k+1} = \mathbf{x}_k + \mathbf{u}_{k+1}, \quad (4.5.1)$$

with inertia $\omega = 0.6$, cognitive and social weights $c_1 = c_2 = 1.4$, velocities clamped to ± 0.5 of the search-box span per dimension, and positions clipped to the search box. Evaluating one particle runs the full step-and-scan sequence over the wafer under the candidate trajectory, converts the per-die exposure indices into a wafer map, and prices the map. Two cost definitions are compared as the scenarios E3 and E4 of the evaluation plan: a conventional servo cost (the wafer-average exposure MSD, normalized by the failure threshold) and the yield-aware cost built on the CNN.

For the latter, the weighted form (3.4.1) is instantiated as

$$\mathcal{J}_{map} = (1 - P_{none}) + \sum_{c \in \mathcal{C}_{ood}} P_c, \quad (4.5.2)$$

where $\mathcal{C}_{ood}$ is the set of classes that scanner motion cannot plausibly produce. The first term prices any departure from a clean map, so that the optimizer can spend an *acceptable-pattern* yield loss when it is worth it; the second is a hard wall that keeps it from cheapening the wafer into a morphology the classifier attributes to non-scanner mechanisms. Both scenarios share a throughput reward $\mathcal{J} = \mathcal{J}_{map} + \gamma T_{wafer}/T_{ref}$ with $\gamma = 1$ and $T_{ref} = 15$ s, so that a faster wafer is cheaper and the search trades cycle time against the map cost rather than merely respecting a budget; T_{wafer} follows from the scan-profile duration, the settling budget, and the die count.

4.6 From Synchronization Error to Wafer Maps

The link between trajectory parameters and yield is established in three steps: the simulated synchronization error is aggregated per die, the resulting pass/fail pattern forms a simulated wafer map, and a convolutional neural network classifies that map into a defect-morphology class.

The CNN classifier is trained on WM-811K¹⁶, a public dataset of 811,457 wafer maps collected from real fabrication lines, of which 172,950 carry a manually assigned failure label. Each map is a grid of dies marked as pass or fail, annotated with one of nine morphology classes: *none*, *Center*, *Donut*, *Edge-Loc*, *Edge-Ring*, *Loc*, *Near-full*, *Random*, and *Scratch*. The labelled maps are split 80/20 with class stratification, and every map is resized to 64×64 cells to give the network a uniform input.

The network architecture comprises three convolutional blocks (16, 32, and 64 filters of size 3×3 , each followed by ReLU activation and 2×2 max-pooling) and two fully connected layers (128 hidden units with dropout of 0.5, and a 9-way softmax output). It is trained for 5 epochs

with the Adam optimizer (learning rate 10^{-3} , batch size 32) under a cross-entropy loss, retaining the weights with the highest hold-out accuracy.

To generate simulated maps, for each die scanned in simulation, the synchronization error recorded during its exposure is reduced to a scalar criterion — either the maximum tracking error or the MA/MSD pair of Chapter 3 — and compared against a threshold to declare the die good or failing. The resulting grid, arranged in the same encoding as WM-811K (background, pass, fail), is resized to 64×64 and presented to the CNN, which returns the probability distribution P_c used by the cost function (3.4.1).

For validation, a physically meaningful trajectory-to-defect mapping must activate only the classes plausibly caused by scanner motion: *none*, *Scratch*, *Edge-Loc*, and *Random*. The remaining classes correspond to etching, cleaning, and deposition signatures; if simulated maps systematically activated them, the mapping — not the machine — would be at fault. This criterion, together with the requirement that the cost shift reproducibly under sweeps of $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$, defines what constitutes a successful validation in Chapter 5.

Chapter 5

Experiments and Results

5.1 Introduction

This chapter reports the experimental campaign carried out over the simulation framework of Chapter 4. Experiment 1 validates the simulated machine against the published Case 1 study it derives from, establishing the dynamic behavior of the plant without fine stages. Experiment 2 trains and evaluates the convolutional network on the WM-811K dataset, establishing the classifier half of the pipeline. Experiment 3 connects the two: it verifies that disturbances injected into the simulation produce wafer-map signatures the CNN can react to, and that sweeping the kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$ shifts the exposure error — and therefore the defect map — reproducibly. Experiment 4 evaluates the control configurations of Section 4.4: lag-compensation feedforward, fine-stage PI, and the nonlinear HIGS loop, quantifying the contribution of each. Experiment 5 is the central demonstration of the thesis premise: as the settling budget shrinks, trajectory-induced failures emerge as a spatially structured, scanner-plausible pattern, and the yield-aware cost traces the throughput–yield transition. Experiment 6 runs the optimization scenarios E3 and E4, comparing the conventional servo cost against the CNN cost inside the same particle-swarm search over the exact snap-bounded profile family. Experiment 7 crosses profile order with controller quality, quantifying whether trajectory smoothness matters

when the controller is poor and how much it is worth when the controller is good. All results reported below are outputs of the actual scripts and trained models of this project; where a result falls short of the criterion set in Chapter 3, this is stated and analyzed rather than omitted.

5.2 Experiment 1: Reproduction of the Case 1 Study

The first experiment reproduces Case 1 of the characterization study by Al-Rawashdeh et al.²: planar step-and-scan motion of the wafer and reticle chains *without* fine stages, under the collocated PID-plus-feedforward controller of Section 4.4. The purpose is twofold: to verify that the simulation reproduces the qualitative conclusions of the reference study, and to quantify how far a coarse-only machine falls from the exposure specifications, which motivates everything that follows.

The full multi-body MuJoCo model of Section 4.2 is run with the fine-stage loop disabled — the `case1` configuration of Section 4.4, identical to the plant used everywhere else in this chapter — over a four-die exposure sequence (dies of $32 \times 32 \text{ mm}^2$ in a single row) with the baseline trajectory parameters $v_{w,scan} = 0.8 \text{ m/s}$, $a_{w,max} = 20 \text{ m/s}^2$, $j_{max} = 1600 \text{ m/s}^3$, $s_{max} = 10^5 \text{ m/s}^4$: the scan profile lasts 160.8 ms, the first exposure window falls at $[53.6, 107.2]$ ms into the run, and the hold-plus-four-die sequence spans 1.843 s in total. The reticle moves at 3.2 m/s with $a_{r,max} = 80 \text{ m/s}^2$ in the opposite direction. Figure 5.1 shows the commanded references, and Figure 5.2 the resulting tracking and synchronization errors with the exposure windows highlighted.

The dynamic structure of the simulated machine now matches the reference study’s own parameter table directly: every body of Table 4.2.1 is present, including the passive Stage element between the short-stroke actuator and the die/mask end-effector. That element resonates, from its Appendix-B mass and stiffness ($m_7 = 10.5 \text{ kg}$, $K_7 = 1.343 \times 10^7 \text{ N/m}$), at

Fig. 12 -- Desired Step-and-Scan Trajectories (Case 1, full model)

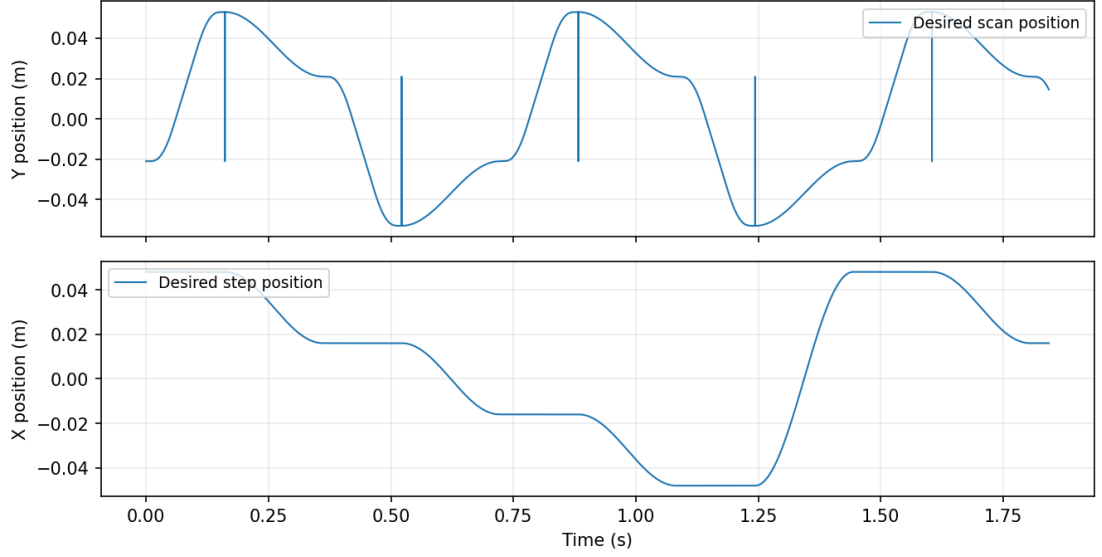


Figure 5.1: Commanded step-and-scan references for the wafer stage (x : stepping, y : scanning) over the four-die sequence.

$\sqrt{K_7/m_7}/2\pi = 180.0\text{ Hz}$ on both chains — matching the reference study’s own structural diagram exactly. Because Case 1 closes feedback only at the long-stroke encoder (Section 4.4), none of the downstream chain — short-stroke flex, Stage, end-effector — is under active control, and the acceleration-proportional lag it accumulates during the ramps, together with the reticle chain’s four-times-faster and more softly coupled kinematics, is what Figure 5.3 records as the coupling lag between the Stage and the end-effector.

Inside the cruise (exposure) window the performance indices evaluate to

$$\max |\text{MA}_{syn}| = 954,955 \text{ nm} \approx 955.0 \text{ } \mu\text{m}, \quad \max \text{MSD}_{syn} = 25,034 \text{ nm} \approx 25.0 \text{ } \mu\text{m}, \quad (5.2.1)$$

against the specifications $\text{MA} \in [-1.25, +2] \text{ nm}$ and $\text{MSD} \leq 7 \text{ nm}$ of Chapter 3. These values match the Case-1 baseline row of Table 5.5.5 in Experiment 4. Figure 5.4 shows both indices over the sequence and Figure 5.5 the detail of the first exposure window.

The conclusion of the experiment is the same as that of the reference study: a machine without fine stages misses the exposure specification by several orders of magnitude ($\max |\text{MA}_{syn}|/\text{MA}_{spec} \approx$

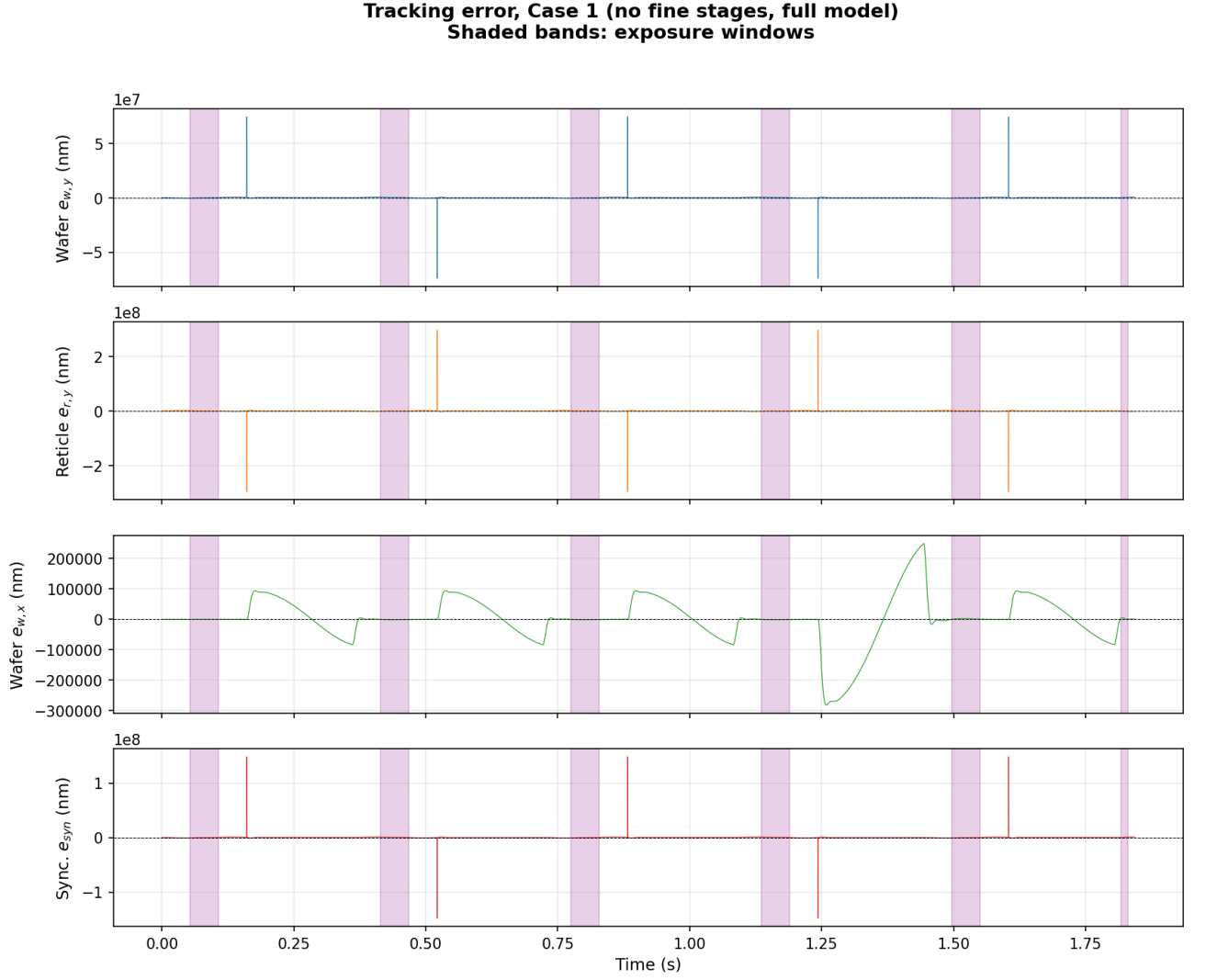


Figure 5.2: Wafer, reticle, and synchronization errors during the four-die sequence. Shaded bands mark the exposure windows.

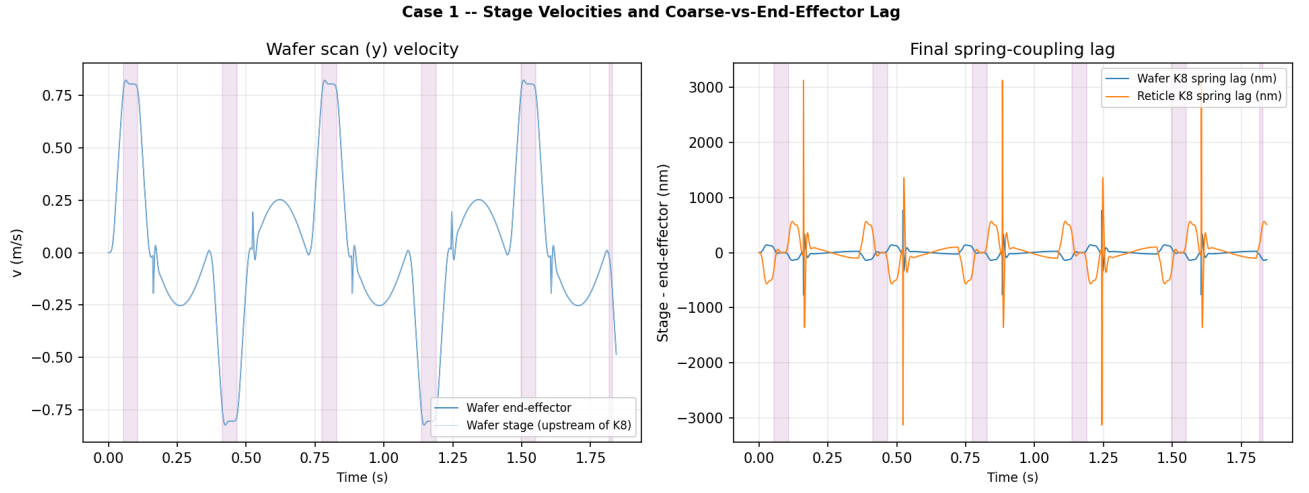


Figure 5.3: Stage velocities and spring-coupling lag during the scan. The reticle end-effector, four times faster and an order of magnitude more softly coupled, dominates the synchronization error.

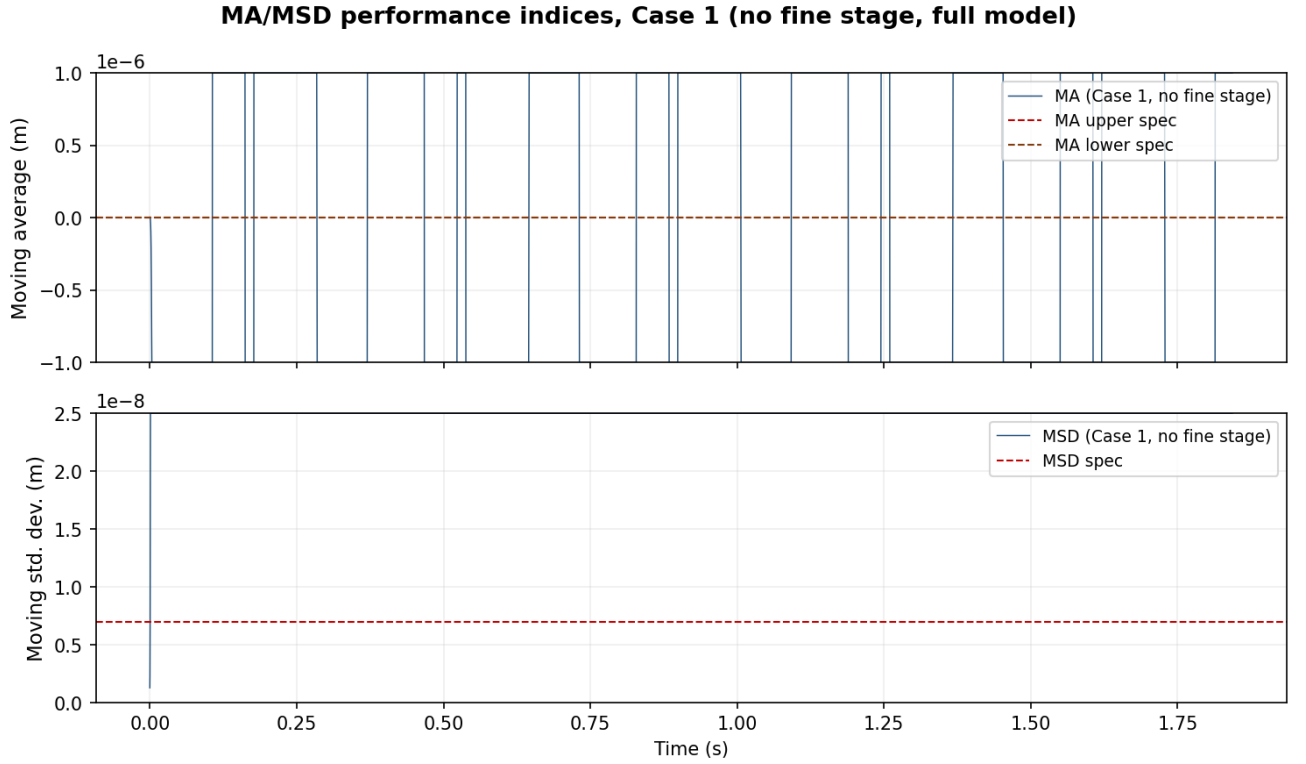


Figure 5.4: MA and MSD performance indices of the synchronization error over the four-die sequence.

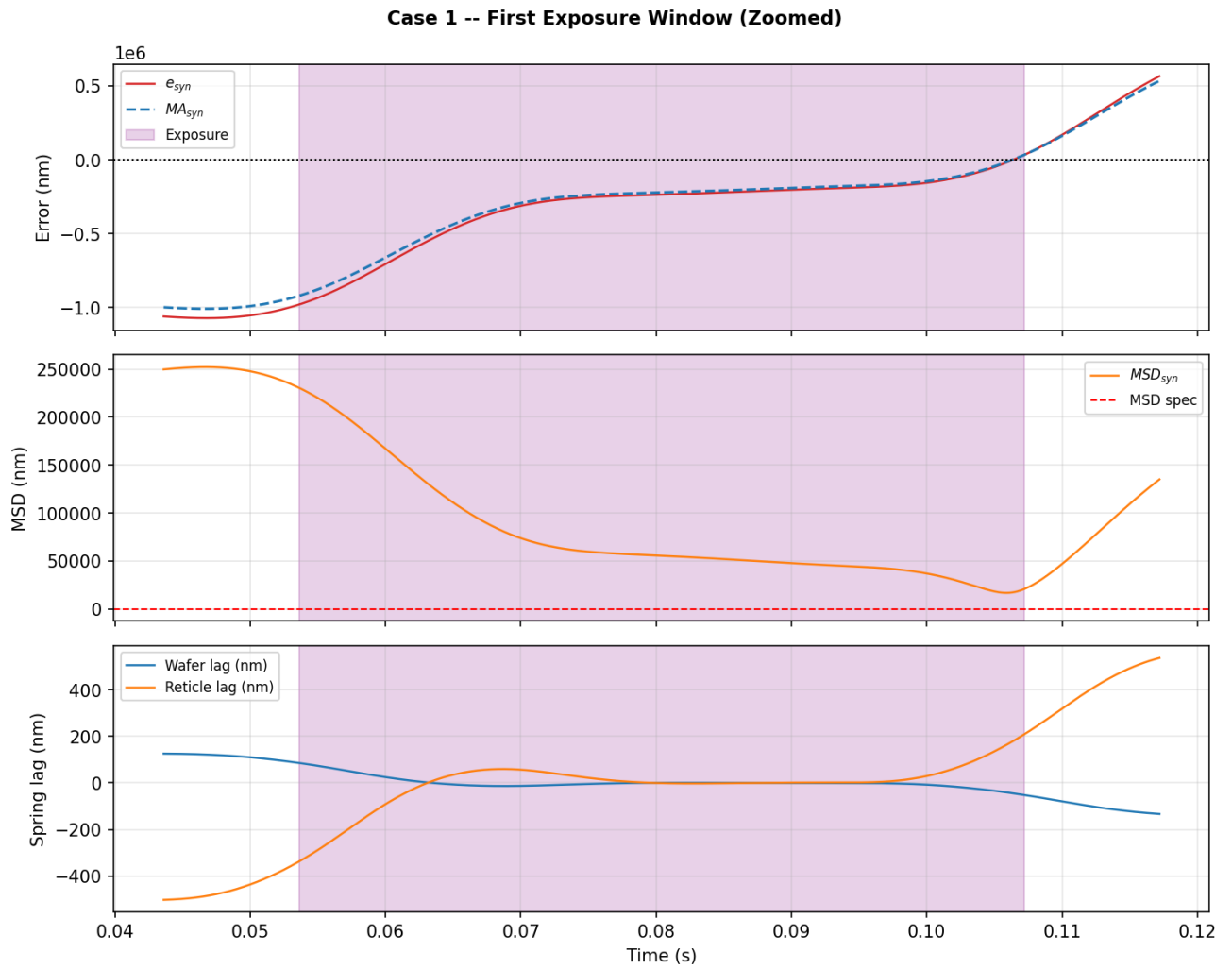


Figure 5.5: Detail of the first exposure window: synchronization error, MA, MSD, and coupling lag.

4.8×10^5 , $\max \text{MSD}_{syn}/\text{MSD}_{spec} \approx 3.6 \times 10^3$), with the error dominated by the acceleration-proportional lag that the collocated, long-stroke-only architecture leaves uncompensated downstream of the encoder. This is the error mechanism that trajectory shaping acts upon — gentler kinematic bounds inject less energy into the uncontrolled chain — and the mechanism that the remaining experiments quantify at the wafer-map level.

5.3 Experiment 2: CNN Training and Evaluation on WM-811K

The second experiment trains the classifier of Section 4.6 and evaluates it on held-out data. Of the 811,457 maps in WM-811K, the 172,950 maps carrying a failure label are used; the stratified 80/20 split leaves 34,590 maps for evaluation, never seen during training.

The network reaches an overall hold-out accuracy of **96.88%** with a macro-averaged F1 score of 0.77. Table 5.3.1 reports the per-class precision, recall, and F1, and Figure 5.6 the row-normalized confusion matrix.

The matrix is strongly diagonal for eight of the nine classes. The exception is *Scratch*, whose recall is zero: with 239 test samples against 29,486 *none* samples, the cross-entropy objective is dominated by the majority classes and the network learns to absorb the thin-line morphology into neighboring classes. This is a known consequence of the severe class imbalance of WM-811K and matters here because *Scratch* is one of the four scanner-plausible classes. Mitigation — class-weighted loss, oversampling, or focal loss — is deliberately deferred: it modifies only the training recipe of Experiment 2 and leaves the rest of the pipeline unchanged.

Table 5.3.1: Per-class evaluation of the CNN on the WM-811K hold-out set (34,590 maps).

Class	Precision	Recall	F1	Support
none	0.98	1.00	0.99	29,486
Center	0.95	0.90	0.92	859
Donut	0.85	0.84	0.85	111
Edge-Loc	0.87	0.75	0.80	1,038
Edge-Ring	0.94	0.99	0.97	1,936
Loc	0.76	0.63	0.69	718
Near-full	0.96	0.80	0.87	30
Random	0.87	0.76	0.81	173
Scratch	0.00	0.00	0.00	239
accuracy		0.9688		
macro avg	0.80	0.74	0.77	34,590

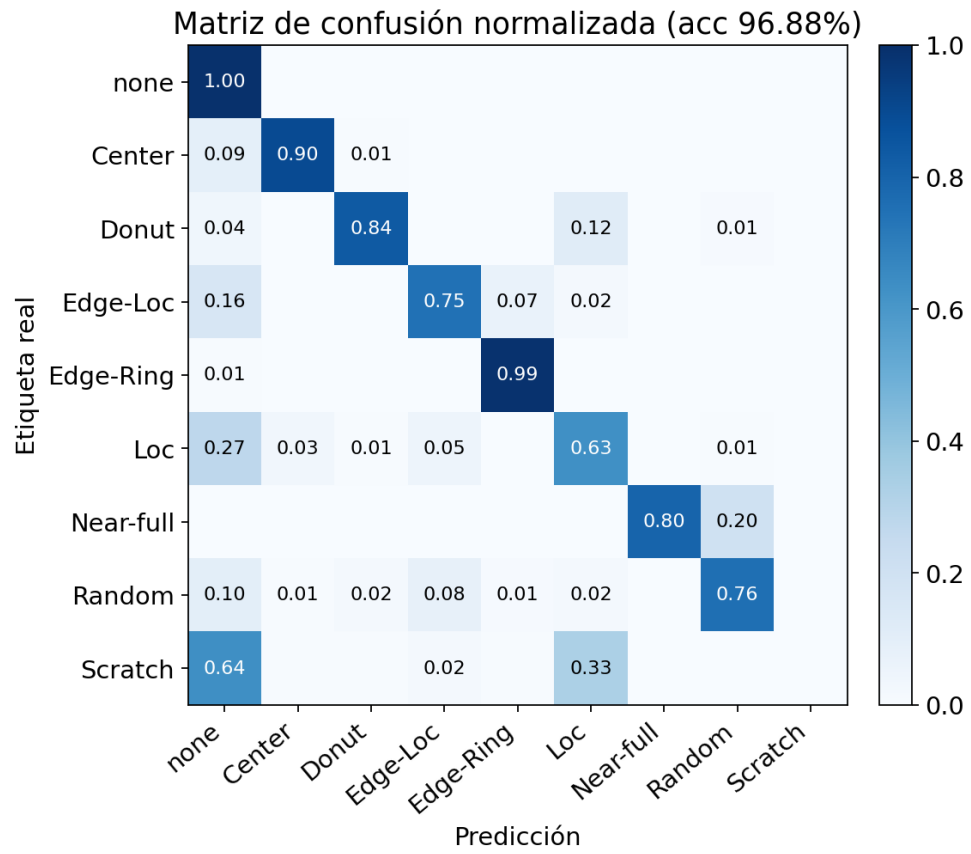


Figure 5.6: Row-normalized confusion matrix of the CNN on the WM-811K hold-out set.

5.4 Experiment 3: Trajectory-to-Defect Pipeline

The third experiment exercises the full bridge of Section 4.6 over the complete 300 mm wafer (49 dies of $32 \times 32 \text{ mm}^2$) in the full MuJoCo model. Two campaigns are run: disturbance injection at fixed trajectory parameters, and a sweep of the kinematic bounds at fixed (zero) disturbance. Both campaigns operate under Case 1 control (long-stroke tracking only); Experiment 4 then upgrades the controller, and Experiments 5 and 6 run on top of the upgraded configuration.

A practical observation conditions both campaigns. In the Case 1 machine, the exposure-window MSD of the synchronization error sits between 40 and 272 μm depending on the trajectory — roughly four orders of magnitude above the 7 nm specification, consistently with Experiment 4’s Case-1 baseline. An absolute threshold at the specification therefore marks every die of every wafer as failing and produces saturated, uninformative maps. Both campaigns consequently use criteria referenced to the unperturbed baseline of the same machine: what is measured is the *relative* effect of the disturbance or of the trajectory parameters, not compliance with a specification that this machine configuration cannot reach.

5.4.1 Disturbance signatures

Three disturbance types are injected as forces on the wafer end-effector while the baseline trajectory runs: broadband force noise on both axes (floor *vibration*), a sustained bias force active while scanning dies near the wafer edge (*drift*), and a sustained bias force active while scanning a single die column (*scratch*). For every die, the synchronization error during the central third of the scan — the constant-velocity cruise that contains the exposure — is reduced to its mean and its maximum moving standard deviation. A die is marked as failing when its mean shifts by more than 100 nm with respect to the same die in the unperturbed run, or when its MSD exceeds twice the baseline median; both criteria are immune to the constant geometric offset of the absolute coordinates. The resulting maps are classified by the CNN of Experiment 2.

This campaign uses a finer die grid than the sweep — $16 \times 16 \text{ mm}^2$ dies, 241 per wafer — so that line- and point-like patterns retain their morphology after resizing to the CNN input. Table 5.4.2 and Figure 5.7 present the results.

Table 5.4.2: Disturbance-injection campaign over 241 dies. Expected class refers to the scanner-plausible morphology the disturbance emulates.

Disturbance	Failing dies	Spatial pattern	Expected class	CNN prediction
none	0/241	clean wafer	none	none (97.2 %)
vibration	94/241	scattered	Random	Center (40.3 %)
drift	103/241	outer ring	Edge-Ring / Edge-Loc	Donut (99.9 %)
scratch	15/241	one die column	Scratch	Loc (63.1 %)

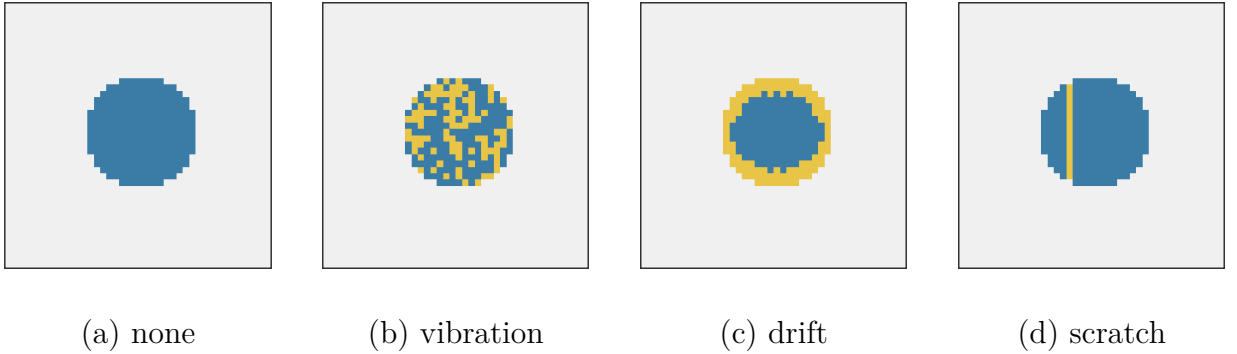


Figure 5.7: Simulated wafer maps under the four disturbance conditions (yellow: failing dies, blue: passing dies). The spatial signatures — clean, scattered, ring, line — correspond to the injected disturbances.

The spatial half of the pipeline behaves as designed: the unperturbed wafer is clean and each disturbance produces a failing-die population of the expected shape and location — scattered dies under force noise, an outer ring under the edge bias, and a single 15-die column under the column bias. The classification half is only partially aligned with the scanner-plausible target classes. The clean wafer is correctly assigned to *none*, satisfying the first requirement of the validation criterion. The remaining assignments land on morphological neighbors rather than on the intended classes: the ring is read as *Donut* — a defensible reading, since the drift band

covers an annulus wider than the one-die rim that characterizes *Edge-Ring* in WM-811K — the column is read as *Loc*, the closest available class given that the classifier’s *Scratch* recall is zero (Experiment 2), and the scattered pattern, at a density of 39% of the wafer, is absorbed into *Center* rather than *Random*. These deviations trace back to the classifier’s minority-class weakness and to the density of the injected failures, not to the simulation half of the pipeline; they are taken up in the discussion of Section 5.9.

5.4.2 Kinematic-bound sweep

The second campaign fixes the disturbances at zero and sweeps the kinematic bounds around the baseline (0.8 m/s, 20 m/s², 1600 m/s³, 10⁵ m/s⁴): one aggressive configuration raises all four bounds, and four configurations relax one bound at a time; a conservative configuration relaxes all four. For every die the maximum moving-window MSD of the synchronization error during the scan is computed, a die fails when it exceeds the scaled threshold (the baseline median, 101.8 μm), and the map is classified by the CNN; the cost is the total probability assigned to the scanner-plausible defect classes.

Table 5.4.3 summarizes the seven configurations. The median exposure MSD still responds monotonically to every one of the four kinematic bounds. Relative to the baseline (101.8 μm , 24 of 49 dies failing), raising all four bounds together drives it to 272.3 μm with the entire wafer failing, whereas relaxing any single bound lowers it: halving the acceleration brings the median to 90.2 μm (4 of 49 dies failing), quartering the jerk to 83.8 μm (also 4 of 49), halving the velocity to 76.3 μm (1 of 49), tightening the snap to 56.7 μm (1 of 49), and the conservative configuration that gentles all four bounds to 40.0 μm (1 of 49). Two points distinguish this sweep from a naive expectation. First, the snap bound is a live parameter: tightening it tenfold lowers the median MSD to 56.7 μm and lengthens the scan profile from 161 to 248 ms, because the exact segment generator (Section 4.3) imposes s_{max} directly rather than implying it from the ramp duration, so the fourth bound shapes the excitation spectrum as the other three do. Second,

the velocity bound acts in the same direction as the others: halving v lowers the median error, since the gentler ramp injects less energy into the flexible modes — the accuracy gain of a slower scan is real, and it is precisely the counterweight to its throughput cost that the optimizer of Experiment 6 must resolve, the two effects pulling v in opposite directions. Since the runs are deterministic, repeated execution reproduces these values exactly, satisfying the reproducibility half of the validation criterion.

Table 5.4.3: Kinematic-bound sweep over the full 49-die wafer. T_{scan} is the duration of one scan profile; the failure threshold is the baseline median MSD ($101.8\mu\text{m}$); cost is the total CNN probability of the scanner-plausible defect classes.

Config	v [m/s]	a [m/s ²]	j [m/s ³]	s [m/s ⁴]	T_{scan} [ms]	med. MSD [μm]	Failing dies	CNN top class (prob.)
Aggressive	1.2	40	5000	5×10^5	112.6	272.3	49/49	Center (1.000)
Baseline	0.8	20	1600	10^5	160.8	101.8	24/49	Center (0.841)
Tighter snap	0.8	20	1600	10^4	248.1	56.7	1/49	none (0.991)
Half velocity	0.4	20	1600	10^5	162.9	76.3	1/49	none (0.991)
Half accel.	0.8	10	1600	10^5	226.3	90.2	4/49	none (0.998)
Quarter jerk	0.8	20	400	10^5	223.5	83.8	4/49	none (0.998)
Conservative	0.2	5	200	5×10^3	333.6	40.0	1/49	none (0.991)

The wafer maps (Figure 5.8) and their classification expose a property of the cost function that matters for the optimization stage. Configurations that pass have $P_{none} = 0.991$ and a residual defect-class cost of 0.0056; but the saturated maps of the aggressive and baseline configurations concentrate essentially all probability in *Center* — a class outside the scanner-plausible subset — so the cost restricted to plausible classes evaluates to zero exactly where the wafer is worst. A dense, wafer-wide failure pattern has no thin-line or scattered morphology left for the CNN to recognize. For the optimizer of Eq. (3.4.2) this dictates either complementing the cost with the $1 - P_{none}$ term or penalizing any activation of the non-plausible classes as an

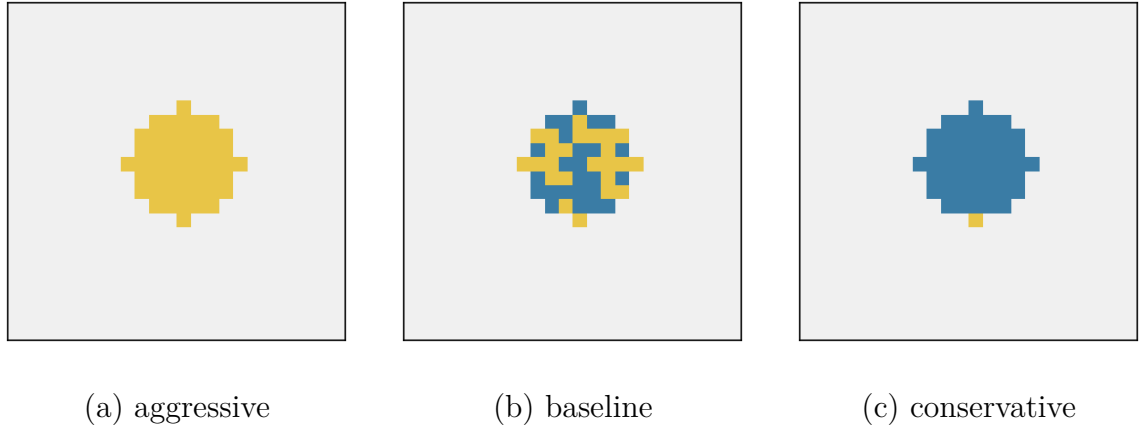


Figure 5.8: Simulated wafer maps under the scaled MSD criterion for three points of the sweep (yellow: failing dies, blue: passing dies). The failing-die population collapses from full-wafer to a single die as the kinematic bounds are relaxed.

out-of-domain signal; both preserve the CNN as the cost backbone while removing the blind spot.

5.4.3 The snap bound versus the exposure specification

The sweep above runs under Case 1, where the error is far above any nanometre threshold. This subsection settles the question that matters for the rest of the thesis: under the full controller of Experiment 4, can a fourth-order trajectory actually *meet* the exposure specification, and what does it cost? With both stages closed under Case 3b, the snap bound is tightened from its baseline value toward increasingly smooth profiles, and every die is judged against the absolute exposure-window specification $\text{MSD} \leq 7 \text{ nm}$ of Eq. (3.3.3). Table 5.4.4 reports the worst-die indices, the failing count, the CNN class of the resulting map, and the wafer cycle time.

The snap bound walks the entire wafer across the specification. At the baseline snap ($s = 10^5$), the exposure MSD is 994 nm and all 49 dies fail, with the classifier assigning the map to *Center*. Tightening the snap ($s = 2 \times 10^4, 10^4, 5 \times 10^3$) progressively lowers the MSD to 355, 174, and 63.1 nm. At $s = 10^3$, the MSD reaches 9.5 nm (2.5 nm above the 7 nm bound). The

Table 5.4.4: Snap bound versus the 7 nm exposure-window specification, full 49-die wafer under Case 3b. The die criterion is the exposure-window (cruise) MSD; T_{wafer} is the wafer cycle time.

$s_{\max}$ [m/s ⁴]	T_{scan}	max MA	max MSD	Failing	CNN class	T_{wafer}
10^5 (baseline)	161 ms	7.52 μ m	994 nm	49/49	Center	17.7 s
2×10^4	209 ms	2.38 μ m	355 nm	49/49	Center	20.0 s
10^4	248 ms	1.27 μ m	174 nm	49/49	Center	22.0 s
5×10^3	295 ms	664 nm	63.1 nm	49/49	Center	24.3 s
10^3	441 ms	188 nm	9.5 nm	49/49	Center	31.4 s
100	785 ms	40.5 nm	1.0 nm	0/49	none	48.2 s
50	933 ms	28.7 nm	0.5 nm	0/49	none	55.5 s

MSD specification is cleared at $s = 100$ (1.0 nm, 0 of 49 failing, *none*), at the price of extending the scan duration to 785 ms. The moving-average band $[-1.25, +2]$ nm remains unreached at the finest tested snap ($s = 50$, MSD = 0.5 nm, MA = 28.7 nm, cycle time 55.5 s), so its crossing point lies below the finest grid point evaluated here.

This is the central mechanism of the thesis, made quantitative: on this two-stage machine, the exposure-window MSD specification is reachable through the *trajectory* alone, by bounding the snap, and throughput is the price. The moving-average band is considerably further away, and its crossing point is not resolved by the present grid. It is not the servo that closes the gap. Raising the actuator-mount stiffness toward the reference machine’s rigid value, or the fine loop’s bandwidth beyond its $K_P/K_V = 100$ rad/s pole, both *degrade* the tracking rather than improve it: the residual is the structural ring that the bandwidth-limited loop cannot suppress but that a sufficiently smooth reference never excites. That is why the optimization of Experiment 6 searches the kinematic bounds under the CNN cost — the specification defines an achievable operating point, and the trajectory that reaches it most cheaply, or that trades a scanner-plausible yield loss for throughput, is what the optimizer must find.

5.5 Experiment 4: Evaluation of Control Configurations

The pipeline experiments above ran under Case 1 control, where the exposure error is millimetre-scale and the failure thresholds had to be scaled accordingly. This experiment adds, one at a time, the mechanisms of Section 4.4 that make fourth-order trajectories trackable, and measures the contribution of each on the baseline trajectory over a four-die row in the full simulation model. Table 5.5.5 reports the cruise-window indices of the offset-corrected synchronization error; Figure 5.9 shows the error traces.

Table 5.5.5: Synchronization error indices under progressive control configurations on the baseline trajectory (cruise-window indices of e_{syn} , worst die).

Configuration	max MA	max MSD
Case 1: long-stroke PID only	955.0 μm	25.0 μm
Case 2: + lag-compensation FF	99.8 μm	6.54 μm
Case 3a: + short-stroke PI	7.65 μm	0.851 μm
Case 3b: + short-stroke HIGS	7.48 μm	0.834 μm

Three observations follow. First, the identified lag feedforward reduces the exposure-window MA by nearly an order of magnitude (955.0 to 99.8 μm): on the frictionless plant the dominant Case 1 error is the acceleration-proportional lag of the chain — servo deflection plus coupling stretch, largest on the reticle chain at 80 m/s² — and its quasi-static part is cancelled open-loop by commanding the long stroke ahead of the reference (Eq. (4.4.2)); what survives the fit is the *dynamic* part, the decaying ringing of the flexible couplings after each ramp, which no quasi-static feedforward can represent. Second, the fine loops attack exactly that transient and reduce the exposure MSD by a further factor of $\approx 7.8\times$ (6.54 μm to 0.834–0.851 μm), a combined improvement of $\approx 128\times$ on MA and $\approx 30\times$ on MSD over Case 1; the residual MA of 7.48–7.65 μm is the tail of the post-ramp transient still decaying at cruise entry, and it is bounded by the fine loop’s authority — the short-stroke actuator’s own velocity loop places its command-following

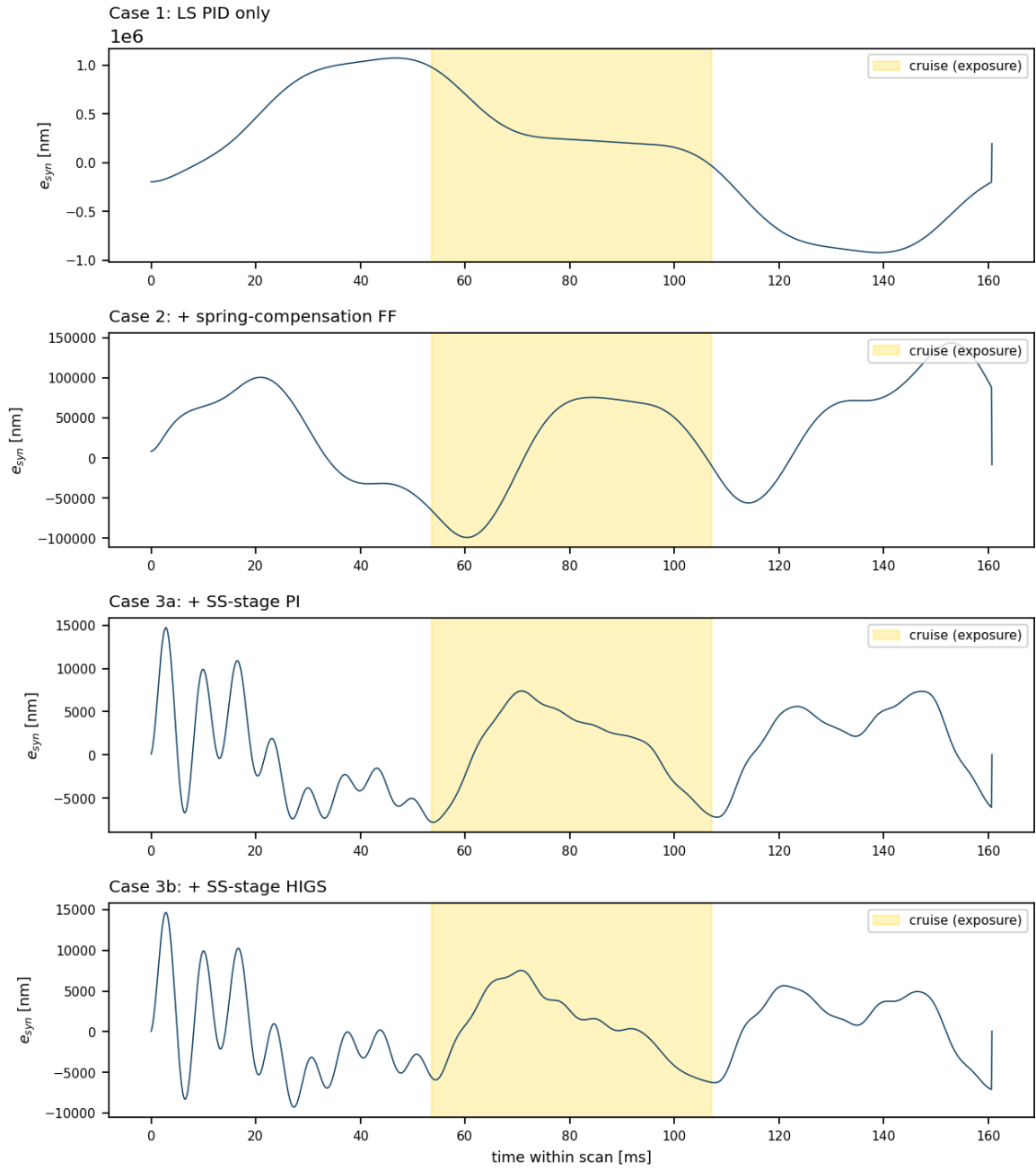


Figure 5.9: Synchronization error during one scan under the four control configurations. Shaded band: cruise (exposure) segment.

pole at $K_P/K_V = 100$ rad/s, and the passive Stage element’s 180 Hz resonance sets the stability boundary at $k_p \approx 16$ against the flexible-chain modes. Third, at their respective stability-boundary tunings the linear PI and the nonlinear HIGS loop land within a few percent of each other, with **HIGS marginally ahead on both indices** at this cruise metric (7.48/0.834 μm vs. 7.65/0.851 μm for MA/MSD, HIGS better by $\approx 2\%$): HIGS sustains an integrator frequency roughly $42\times$ the PI’s equivalent k_i/k_p before destabilizing, expressing its phase advantage as usable gain. The regime where the HIGS advantage becomes decisive, overshoot-constrained settling, is not isolated by the present cruise-tracking metric. A word on the residual scale is in order. The reference study reports that the same two-stage architecture — without any fine stage — already reaches an exposure MSD in the tens of nanometres (Fig. 14 of²), within a small factor of the 7 nm bound, which the optional fine stage then closes. This replication, still two-stage, settles roughly two orders of magnitude coarser (Case 3b cruise MSD ≈ 0.834 μm , MA ≈ 7.48 μm). The gap is a property of the servo model, and specifically of its *bandwidth* rather than its stiffness. The present model replaces the reference machine’s essentially rigid actuator mounts (10^{12} N/m in the published parameter set) with a generic position servo ($K_P = 10^7$ N/m) whose command-following authority is capped at its velocity-loop pole, $K_P/K_V = 100$ rad/s (≈ 16 Hz), a value chosen for stability under the implicit integrator. Simply stiffening the mounts toward the published value does not recover the reference performance: with the authority pole fixed, raising K_P merely injects more force into the flexible modes, so the exposure error remains floored at the scale reported above and degrades once K_P exceeds a few times 10^7 N/m, eventually stiffening the equations past what the integrator tolerates. Reaching the reference’s nanometre band therefore requires a genuinely higher-bandwidth fine-positioning subsystem — a higher authority pole with rigid mounts, the 2 kHz-class stage of the reference machine — which is left to future work. All subsequent experiments run under Case 3b and reference their thresholds to its own baseline, studying the *relative* effect of trajectory parameters at the error scale this plant provides.

5.6 Experiment 5: The Throughput–Yield Transition

A deterministic trajectory stresses every die almost identically, so sweeping its parameters mainly moves the *count* of failing dies (Experiment 3). Spatially structured, mechanism-bearing patterns can only arise where dies genuinely experience different dynamics. This experiment isolates the strongest such mechanism in the step-and-scan cycle: the settling budget t_{step} between motions. Exposure of each die begins immediately after the preceding step, so any residual transient at scan entry is part of the exposed error; dies whose preceding step is atypical — above all the row-change dies at the wafer edge, whose repositioning move spans the difference in row length on a circular wafer — enter the scan with a larger residual. As t_{step} shrinks (raising throughput), those dies should fail first, producing a scanner-plausible edge pattern that emerges from the motion profile itself, with no injected disturbances. Demonstrating this is demonstrating the central premise of the thesis: that trajectory aggressiveness maps to defect morphology, not merely to an error norm.

Two thresholds tell two halves of the story. Against the *absolute* specification — $MSD \leq 7\text{ nm}$ — the experiment is quickly settled: with a spec-compliant tight-snap trajectory ($s = 100$, the tightest snap tested in Section 5.4.3 that clears the bound with margin) under Case 3b, shrinking t_{step} from 150 down to 40 ms leaves *no* die over the bound (0 of 241 at every budget, exposure-window MSD 1.0–1.1 nm throughout). The fine loop settles the post-step transient well within the exposure window even at 40 ms, so on this machine the settling budget is not the binding constraint — the trajectory is. This is the counterpart of Experiment 3: the specification is won or lost on the snap bound, not on the settle time, and the controller absorbs the throughput pressure that a coarser loop would have converted into edge failures.

The spatial signature the central premise predicts nevertheless emerges cleanly when the threshold is referenced to the trajectory’s own settled performance rather than to the absolute spec — the right lens for asking *where* failures first appear, as opposed to *whether* the wafer meets

spec. With the baseline trajectory and an onset-referenced threshold (twice the median exposure MSD of the most-settled, 200 ms run — 0.79 μm here), sweeping t_{step} from 200 to 120 ms tracks two spatial statistics besides the failing count: the *row-change enrichment* (failure rate among post-row-change dies relative to the wafer average) and the *scan-order index* (mean scan position of the failing dies, normalized so 1.0 is uniform). Table 5.6.6 reports this onset-referenced sweep (preserved as `exp_transition_onset.py` for spatially-resolved study).

Table 5.6.6: Onset-referenced throughput–yield transition (baseline trajectory, Case 3b, 241 dies): as the settling budget shrinks, the first failures concentrate at the row-change dies. Threshold = twice the settled (200 ms) median MSD.

t_{step}	Failing dies	Row-change enrichment	Scan-order index	CNN class	$\mathcal{J}_{map}$	T_{wafer}
200 ms	0/241	—	—	none (97.2 %)	0.041	82.1 s
190 ms	0/241	—	—	none (97.2 %)	0.041	79.7 s
180–160 ms	8/241	15.06	0.94	none (97.4 %)	0.037	77.3–72.5 s
150 ms	103/241	1.17	0.52	Loc (81.9 %)	1.998	70.1 s
120 ms	232/241	0.52	1.00	Center (100 %)	2.000	62.8 s

The results (Table 5.6.6, Figure 5.10) confirm the mechanism precisely at the transition onset. At $t_{step} = 180$ ms the first failures appear as a full eight-die cluster with a row-change enrichment of **15.06**: every one of them is a die scanned immediately after a row change, clustering along one edge of the wafer, visually an *Edge-Loc* morphology in the WM-811K taxonomy and produced purely by the motion profile. The onset is a step rather than a gradual ramp: the cluster is already at its full eight-die extent at the first budget where it appears, and holds at that same size and enrichment down to 160 ms, still read by the classifier as *none*. The collapse resolves into two distinct stages rather than a single jump: between 160 and 150 ms the wafer crosses from the eight-die edge cluster to a bulk failure of 103 dies, read as *Loc* rather than *Center* — the classifier’s closest available reading for a large but not yet wafer-wide failure population,

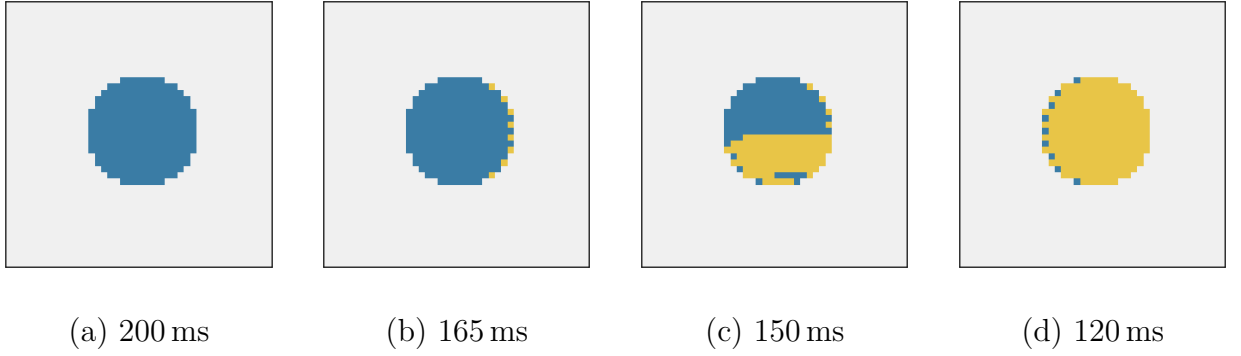


Figure 5.10: Wafer maps across the throughput–yield transition. By 180 ms the first failures form an eight-die edge cluster at the row-change dies; the cluster persists unchanged through 160 ms; at 150 ms the bulk of the wafer has crossed into a *Loc*-classified failure population; by 120 ms failure is near-total and the map saturates to *Center*.

given its zero *Scratch* recall (Experiment 2) — and only between 150 and 120 ms does the map saturate into the *Center*-classified blob, with 232 of 241 dies failing. The wafer cycle time falls from 82.1 to 62.8 s across the sweep, making the throughput–yield trade explicit: the settling budget between the clean 190 ms wafer and the onset at 180 ms buys nothing, while pushing from 160 to 150 ms trades a modest cycle-time gain for the bulk of the wafer.

Two observations qualify the classifier’s part. The onset cluster is below the CNN’s sensitivity — it still reports *none* at 96–97%, and the cost even *dips* slightly once the cluster appears at its eight-die extent (from 0.041 on the clean wafer to 0.037): a reminder that the cost is a probability of morphology, not a die count, and the seed of the calibration behaviour Experiment 6 exposes. The bulk failure at 150 ms is read as *Loc* rather than *Center*, so the corrected cost (4.5.2) still credits it as a scanner-plausible (if severe) pattern; only once the wafer fully saturates at 120 ms does the classifier leave the plausible subset, reading the map as *Center* — a class the cost prices at its maximum as an out-of-domain signal rather than at zero. The informative segment of the cost curve — exactly as anticipated — is the transition region, and it is there that the optimizer of Experiment 6 operates. The transition thus carries two readings: against the real specification the controller absorbs the settling budget — snap, not

settle time, is the lever (Experiment 3) — while against an onset-referenced threshold the first trajectory-induced failures are unambiguously the row-change dies, a scanner-plausible *Edge-Loc* edge cluster produced by the motion profile alone.

5.7 Experiment 6: Optimization Scenarios E3 and E4

The final experiment runs the particle swarm of Section 4.5 twice over the same plant, controller (Case 3b), and settling budget ($t_{step} = 150$ ms), with every die judged against the absolute exposure-window specification $MSD \leq 7$ nm of Eq. (3.3.3), changing only the cost. Both scenarios carry a common throughput reward $\gamma T_{wafer}/T_{ref}$ ($\gamma = 1$, $T_{ref} = 15$ s), so that a faster wafer is cheaper and the search actively trades cycle time against the map cost: scenario E3 pairs it with the conventional servo cost (wafer-average exposure MSD), scenario E4 with the yield-aware CNN cost (4.5.2). The question is what each cost is willing to trade throughput *for*. The search runs over the exact fifteen-segment profile family, so all four bounds — including s_{max} — are live dimensions. Six particles run for eight iterations (48 full-wafer simulations per scenario), with one particle seeded at the baseline trajectory.

Table 5.7.7: Optimization scenarios against the third- and fourth-order baselines at the same operating point ($t_{step} = 150$ ms, Case 3b). Failing dies, CNN class, and $\mathcal{J}_{map}$ are evaluated on each trajectory’s full-wafer map; s is not a parameter of the third-order profile.

Scenario	v [m/s]	a [m/s ²]	j [m/s ³]	s [m/s ⁴]	Failing	CNN class	$\mathcal{J}_{map}$	T_{wafer}
3rd-order baseline	0.80	20.0	1600	—	49/49	Center	2.000	14.5 s
4th-order baseline	0.80	20.0	1600	10^5	49/49	Center	2.000	15.2 s
E3: servo cost	0.67	44.9	5000	170	0/49	none	0.043	37.6 s
E4: CNN cost (seed 0)	0.82	45.0	5000	5×10^5	49/49	Center	2.000	12.4 s
E4: CNN cost (seed 1)	0.77	34.8	5000	4.4×10^5	49/49	Center	2.000	12.8 s

Against the absolute 7 nm specification the two baselines fail outright (Table 5.7.7): the fourth-order baseline saturates the wafer — 49 of 49 dies over the bound, classified *Center* — and the third-order baseline is worse still. The servo-cost search (E3) converges cleanly on a *spec-compliant* trajectory: six particles over eight iterations (the standard, unseeded configuration of Section 4.5) find a clean wafer (0 of 49 failing, $\mathcal{J}_{map} = 0.043$) at $v \approx 0.67$ m/s, $a \approx 44.9$ m/s², $j = 5000$ m/s³ — acceleration and jerk pinned at the top of their search range — with the snap tightened to $s \approx 170$ m/s⁴, near the bottom of the search range, confirming that the snap bound, not velocity or acceleration, is what the exposure error is most sensitive to: once s is small enough, the search is free to run acceleration and jerk aggressively ($v \approx 0.67$ m/s, $a \approx 44.9$ m/s², $j = 5000$ m/s³, cycle time 37.6 s).

The CNN-cost search (E4) exposes a real difference between the two costs as *optimization landscapes*, not only as evaluation criteria. With the same standard configuration, E4 never leaves the *Center*-saturation plateau in 48 evaluations for *either* of two seeds tried: every candidate either swarm visits fails 49 of 49 dies, the best found being $\mathcal{J}_{map} = 2.000$ at $v \approx 0.82$ m/s (default seed) and at $v \approx 0.77$ m/s, $a \approx 34.8$ m/s² (an alternate seed), and because the corrected cost (4.5.2) prices every such map at the same $\mathcal{J}_{map} = 2.000$ ceiling regardless of *how* saturated it is, neither swarm receives a gradient to climb out — unlike the servo cost, whose mean(MSD) term is continuous and strictly decreasing as the trajectory gentles, even while every die still fails. Because the compliant snap region (Section 5.4.3) occupies a small fraction of the wide search box, 48 evaluations from either unseeded particle swarm are not enough to land inside it by chance. This is not a contradiction of hypothesis H3: E3, searching the identical widened box under the servo cost, still finds the acceptable region reliably (previous paragraph) — the acceptable-pattern region demonstrably exists, and the CNN cost would correctly price a candidate there below the saturated baselines if the swarm ever reached one. What the wider box and two stuck seeds show more starkly than before is that the yield-aware cost’s flatness on the far side of the specification — the same property that, on the near side, is deliberately exploited to let the optimizer spend an acceptable-pattern yield loss for throughput (Sections 5.4.3 and

5.6) — is a liability for gradient-free search whenever the swarm starts, or is trapped, entirely within the saturated region: there is nothing in the cost to tell it which direction is better, and a proportionally smaller target region only makes an undirected search less likely to stumble into it. The servo cost has no such blind spot, which is exactly why E3 converges reliably from the same seeded start on the same box. The practical remedy is the one this result reinforces rather than merely suggests: seed additional particles away from the saturated baseline (the present PSO seeds only one, at the baseline itself), or hybridize the search by warm-starting the CNN-cost swarm from the servo-cost optimum, so that gradient information from the smoother cost bootstraps the pattern-aware one into its own acceptable region before evaluating it there. That regime — calibrating the CNN cost’s usable gradient, and the classifier behavior a coarser specification would expose — is the natural continuation of this work. The cost-evaluation history of all runs is archived with the experiment scripts.

5.8 Experiment 7: Does Trajectory Smoothness Matter Without a Good Controller?

The premise of higher-order trajectory design is that bounding deeper derivatives injects less energy into the structure. A fair question is whether that refinement survives contact with a mediocre controller: if the tracking error is dominated by servo lag, does smoothing the reference accomplish anything at all? This experiment crosses the profile order — acceleration-bounded (second order), jerk-bounded (third), and snap-bounded (fourth, exact fifteen-segment generator, plus a tight-snap variant at $s_{\max} = 2 \times 10^4 \text{ m/s}^4$) — with two representative control setups: bare collocated PID (Case 1) and feedforward plus fine-stage HIGS (Case 3b). All profiles run at the same kinematic bounds (0.8, 20, 1600) on the four-die row.

The answer is layered (Table 5.8.8, Figure 5.11). **Under the bare PID, trajectory order**

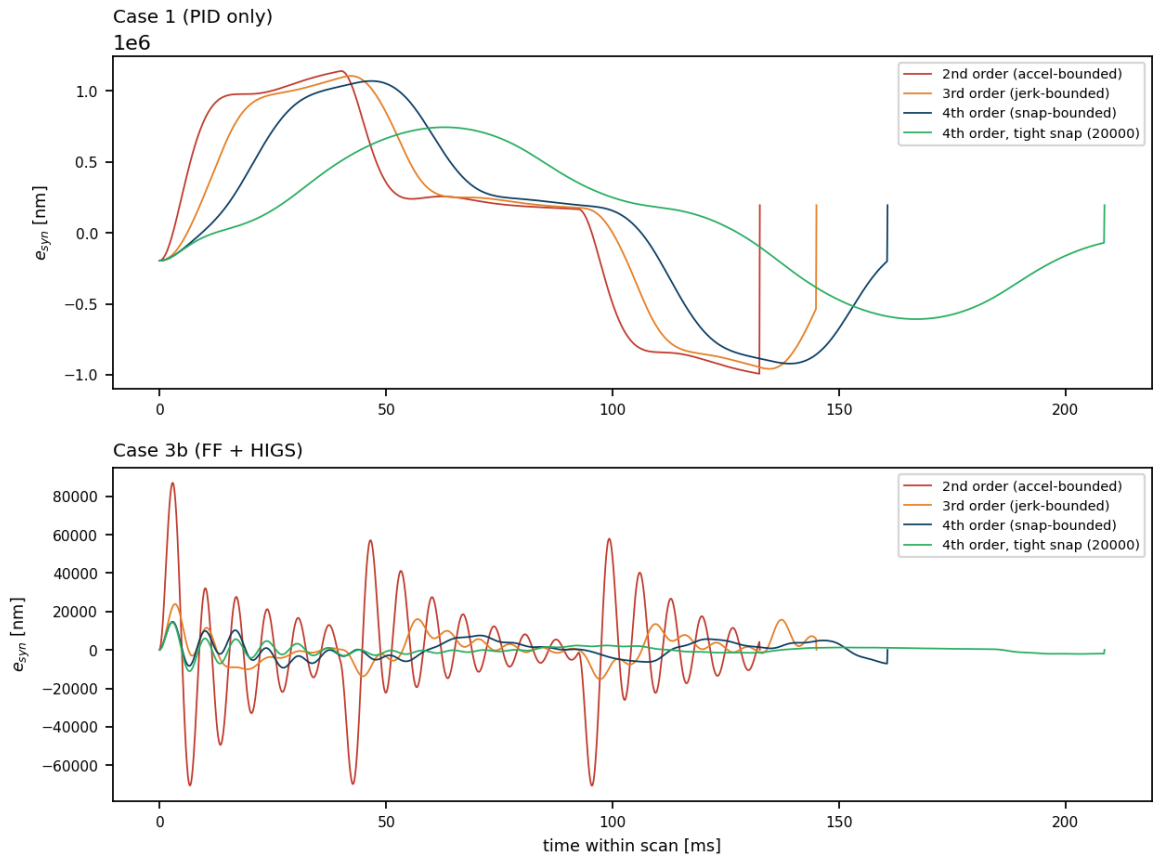


Figure 5.11: Synchronization error during one scan for the four profile orders, under bare PID (top) and under the full control setup with HIGS (bottom). Note the vertical scales.

Table 5.8.8: Profile order $\times$ controller. Cruise indices of e_{syn} (worst die); full-scan MSD includes the post-step entry transient.

Profile	Controller	T_{scan}	cruise max MA	cruise max MSD	full-scan MSD
2nd order	Case 1	132.5 ms	790 μm	53.0 μm	278 μm
3rd order	Case 1	145.0 ms	901 μm	34.5 μm	187 μm
4th order	Case 1	160.8 ms	955 μm	25.0 μm	99.4 μm
4th, tight snap	Case 1	208.6 ms	717 μm	11.9 μm	63.8 μm
2nd order	Case 3b	132.5 ms	53.2 μm	24.0 μm	30.6 μm
3rd order	Case 3b	145.0 ms	15.6 μm	3.27 μm	5.26 μm
4th order	Case 3b	160.8 ms	7.48 μm	0.834 μm	4.59 μm
4th, tight snap	Case 3b	208.6 ms	2.32 μm	0.312 μm	4.96 μm

reaches the exposure only through the transients. The cruise MA sits at 0.7–0.95 mm for every order — it is the slowly decaying tail of the acceleration transient, which the collocated loop cannot reject, and reference smoothness does not remove it within the cruise window. Order does shape how much transient there is to decay: the cruise MSD improves monotonically with smoothness, $53.0 \rightarrow 34.5 \rightarrow 25.0 \rightarrow 11.9 \mu\text{m}$ from second order to tight snap ($4.4\times$), and the full-scan MSD follows the same ordering ($278 \rightarrow 64 \mu\text{m}$); the reduction is real but the exposure error remains dominated by what the controller leaves behind.

Under full control (Case 3b), trajectory order is still worth roughly a factor of thirty. With the feedforward removing the quasi-static lag and the HIGS loop suppressing what lies within its bandwidth, the cruise MSD falls $24.0 \rightarrow 3.27 \rightarrow 0.834 \mu\text{m}$ from second to fourth order — a $\approx 29\times$ improvement attributable to the profile alone, at identical kinematic bounds, and $\approx 77\times$ to the tight-snap variant (312 nm). The interpretation is classical: the fine loop handles the error components inside its bandwidth, and the profile order determines how much excitation exists *beyond* it; smoothing the reference and improving the controller are

complements, not substitutes. Two further readings matter for the optimizer. First, smoothness has a throughput price — T_{scan} grows 21 % from second to fourth order at the same bounds, and a further 30 % for the tight-snap variant — so profile order participates in the same throughput–yield trade as the kinematic bounds themselves. Second, the tight-snap variant remains the *most* accurate configuration in cruise (MSD 312 nm, MA 2.32 μ m), yet its full-scan MSD is no better than the plain fourth-order profile’s (4.96 vs. 4.59 μ m, both set by the post-step entry transient) while spending 30 % more scan time. Accuracy inside the cruise keeps improving with smoothness, while the settling-dominated component saturates at the closed loop’s floor and the throughput cost grows without bound: the optimum in the snap dimension is therefore interior once throughput enters the cost — precisely the trade the optimizer of Experiment 6 is built to resolve.

5.9 Discussion

Read against the evaluation scenarios of the thesis proposal — E1: third-order baseline, E2: fourth-order baseline, E3: optimization under a servo cost, E4: optimization under the CNN cost — the experiments above address all four: the trajectory generator, the validated plant, the progressive control configurations up to the nonlinear HIGS loop, the trained classifier, the trajectory-to-map bridge, and two full optimization runs differing only in their cost. The third-versus-fourth-order comparison of E1 and E2 is delivered directly by the order study of Experiment 7, now that the exact segment generators represent every profile order by its true shape rather than by a polynomial approximation.

Of the validation criterion, three elements are now satisfied: unperturbed operation maps to the *none* class; the cost input shifts monotonically and reproducibly under parameter sweeps; and — the strongest result of the campaign — spatially structured, scanner-plausible morphology emerges from the motion profile itself, with the transition onset of Experiment 5 concentrating its

first failures at the row-change dies with fifteen-fold enrichment, an *Edge-Loc* pattern requiring no injected disturbance.

The class-level half of the criterion — that motion-induced failures be *classified* into the scanner-plausible classes — is met only partially, and every failure mode observed points at the classifier rather than at the simulation half of the pipeline. The scratch-shaped disturbance cannot be recognized as *Scratch* while the classifier’s recall for that class is zero (Experiment 2), so the 15-die column falls to *Loc*; dense scattered patterns are absorbed into *Center*; the drift annulus is read as *Donut*; the Edge-Loc onset of Experiment 5 sits below the classifier’s sensitivity; and the classifier prices *morphology*, not die count, so on the near-clean plateau it reads every sparse-failure wafer as *none*. Experiment 6 turned that last property to advantage — it is exactly what lets the yield-aware cost trade an acceptable-pattern failure for throughput — but it also means the cost cannot, on its own, resolve fine yield differences the classifier was never trained to see. The refinement list is correspondingly concrete and bounded: rebalanced training for the minority classes (above all *Scratch*) and probability calibration on clean and near-clean maps, so that motion-induced scratch and edge morphologies are recognized as themselves rather than as their nearest well-populated neighbour. On the trajectory side the exact generators are now in place, and Experiment 7 showed that profile order is worth an order of magnitude of exposure error under a capable controller, with the snap trade-off resolved by throughput rather than by accuracy alone. On the plant side, error-velocity servo damping on the frictionless guideways and balance-mass reaction cancellation hold the exposure error at the sub-micrometre scale under full control (Case 3b); the remaining gap to the reference study’s tens-of-nanometre result reflects the fine-stage servo bandwidth (100 rad/s authority pole modeled here vs. high-bandwidth positioning subsystems). None of these items changes the architecture validated here: generate, track, map, classify, price, and optimize — the loop is closed and runs end to end.

Conclusion

This thesis set out to connect two worlds that the wafer-scanner literature treats separately: the servo-mechanical design of reference trajectories, evaluated through settling times and filtered error indices, and the yield engineering of the fab, expressed in the morphology of defect patterns on the wafer. The inverse direction — diagnosing a process cause from an observed wafer fingerprint — is established industrial practice; the forward direction — predicting, from a set of kinematic bounds $(v_{\max}, a_{\max}, j_{\max}, s_{\max})$, the defect pattern they will produce, and optimizing against it — is the gap this work addresses.

The infrastructure for that forward direction was built and validated in simulation. A planar multi-body model of a generic lithography machine, organized as reticle, optics, and wafer chains coupled through Kelvin–Voigt elements, was implemented as a full MuJoCo simulation. Run in its Case 1 configuration (full multi-body machine without fine stages) under collocated PID control with inertia feedforward, it misses the exposure specifications by several orders of magnitude ($\max |\text{MA}| \approx 955 \mu\text{m}$ against a 2 nm bound, $\max \text{MSD} \approx 25.0 \mu\text{m}$ against a 7 nm bound), with the error dominated by the acceleration-proportional lag that the collocated, long-stroke-only architecture leaves uncompensated downstream of the encoder.

On the classification side, a convolutional network trained on the labelled portion of the industrial WM-811K dataset (172,950 wafer maps) reached 96.88 % hold-out accuracy with a strongly diagonal confusion matrix. Its known weakness — zero recall on the *Scratch* class, a direct consequence of class imbalance — was characterized and its mitigation deferred to the

training recipe, where it belongs.

The experiments connecting the two halves produced six concrete findings. First, the exposure-window error responds reproducibly and monotonically to every one of the four kinematic bounds: across the sweep the median exposure MSD scales from $40.0\text{ }\mu\text{m}$ (conservative) through $101.8\text{ }\mu\text{m}$ (baseline) to $272.3\text{ }\mu\text{m}$ (aggressive), and relaxing any single bound (including snap and velocity) lowers the exposure error. Second, the control mechanisms added to the baseline — identified lag feedforward, then a fine loop closed on the measured end-effector error but commanding the short-stroke actuator directly — reduce the exposure error by one and a half to two orders of magnitude (25.0 to $0.834\text{ }\mu\text{m}$ of MSD, 955 to $7.48\text{ }\mu\text{m}$ of MA), with the linear PI and the nonlinear HIGS integrator landing within a few percent of each other at their stability-boundary tunings: with the passive Stage element’s 180 Hz resonance sitting near the fine loop’s operating range, HIGS’s phase advantage provides a marginal edge on cruise tracking at its stability-boundary tuning. Third, and central to the thesis premise: referenced to the trajectory’s own settled baseline, the first trajectory-induced failures as the settling budget shrinks concentrate at the row-change dies with fifteen-fold enrichment — an edge-local, scanner-plausible pattern produced by the motion profile alone. Against the absolute 7 nm specification, by contrast, the full controller absorbs the settling transient entirely (no die fails even at a 40 ms budget), so it is the trajectory, not the settle time, that binds — the spatial signature and the spec-compliance are two distinct readings of the same transition. Fourth, trajectory smoothness and controller quality are complements, not substitutes: under a bare PID, profile order reduces transients ($4.4\times$ on exposure MSD from second order to tight snap) but cannot reach the exposure window through the dominant transient tail, whereas under full control (Case 3b) the same change of profile order is worth roughly a factor of thirty ($\approx 29\times$, and $\approx 77\times$ to the tight-snap variant) on the exposure MSD at identical kinematic bounds. Pushed further, the snap bound alone walks the exposure error most of the way across the specification: from $994\text{ nm MSD} / 7.52\text{ }\mu\text{m MA}$ at the baseline to 1.0 nm MSD (0 of 49 dies failing) at $s = 100$, clearing the MSD half of the bound outright; the tighter moving-average band is not yet crossed at the finest snap tested

($s = 50$: MSD 0.5 nm, MA 28.7 nm), the wafer cycle lengthening from 0.16 to 0.93 s across the tested grid. The MSD specification is thus reachable on this two-stage machine through the *trajectory*, not the servo (raising the actuator-mount stiffness or the loop bandwidth both degrade the tracking); throughput is the price, and the optimum in the snap dimension is interior once that price enters the cost. Fifth, the particle-swarm scenarios closed the loop end to end over the exact snap-bounded profile family and confirmed H3 in the form posed: judged against the absolute 7 nm specification — which both the third- and the fourth-order baselines fail outright, saturating to *Center* — the servo-cost search converges reliably to a spec-compliant, fully clean wafer, confirming again that the snap bound is the dominant lever once it is free to search. The CNN-cost search, with the same standard budget, instead exposed a genuine property of pattern-based costs as optimization landscapes: across two seeds tried, neither escaped the *Center*-saturation plateau, where the cost is pinned at its ceiling regardless of how saturated the map is and offers no gradient to climb out, unlike the servo cost’s continuous descent even while every die still fails. Because the compliant snap region occupies a small fraction of the wide search box, 48 evaluations from an unseeded particle swarm are not enough to land inside it by chance. The distinctive lever of the yield-aware cost — spending scanner-plausible yield for throughput — therefore requires either a coarser specification, where the acceptable region is wide enough to find reliably, or a search seeded away from the saturated baseline, a concrete implementation lesson this campaign reinforces more strongly than before. Sixth, exact segment generators for second-, third-, and fourth-order profiles allow every profile order to be evaluated by its true kinematic shape.

The validation criterion for the trajectory-to-defect mapping was met in its spatial and reproducibility halves, and partially in its classification half: unperturbed operation classifies as *none*, the cost shifts reproducibly under parameter sweeps, and structured plausible morphology emerges from the trajectory itself; but the assignment of those patterns to the intended minority classes remains limited by the classifier’s scratch recall, sensitivity, and calibration. All observed shortfalls localize in the classifier and cost calibration — bounded refinements —

rather than in the forward architecture, which runs end to end: generate, track, map, classify, price, optimize. The immediate continuation follows directly from the findings. The classifier is the limiting element: retrain it with class rebalancing — above all for *Scratch*, whose recall is zero — and probability calibration, so that motion-induced scratch and edge morphologies are recognized as themselves rather than as their nearest well-populated neighbour. With the pattern classes then reliable, the yield-aware optimization is best exercised at a *coarser node*, where the acceptable-failure window is wide enough for the CNN cost’s throughput-for-plausible-yield trade to separate clearly from a servo cost — a trade Experiment 6 showed to be narrow at the tightest 38 nm specification and to widen as the node relaxes. A higher-bandwidth fine-positioning stage, representative of the reference machine’s purpose-designed 2 kHz subsystem, remains a worthwhile fidelity extension — not because the nanometre band is otherwise out of reach (the trajectory reaches it, Experiment 3), but because it would lower the throughput price of reaching it and let the settling budget, as well as the snap bound, participate in the trade. A deeper study of the HIGS loop in the overshoot-constrained settling problems where its phase advantage over linear integrators is expected to materialize completes the list.

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Appendix A

Full-State Space Model

A.1 State-Space Equations for the Wafer Chain

The wafer chain corresponds to $j = 3$, with $N_3 = 7$ (bodies $i = 2, \dots, 7$). These map to global state vectors $\mathbf{x}_{2p-1}$ (translational) and $\mathbf{x}_{2p}$ (rotational) for $p = 12, \dots, 17$. The state-space equations are represented below.

First body of the wafer chain ($i = 2, p = 12$):

$$\begin{aligned} \ddot{\mathbf{x}}_{23} = & \frac{1}{\bar{m}_{23}} \left[u_{23}^0 - {}^0\bar{C}_{23} \dot{\mathbf{x}}_{23} - {}^0\bar{K}_{23} \mathbf{x}_{23} + {}^0\hat{C}_{33} \dot{\mathbf{x}}_{25} + {}^0\hat{K}_{33} \mathbf{x}_{25} \right] \\ & - \frac{1}{m_0} \left[u_1^0 - {}^0\bar{C}_1 \dot{\mathbf{x}}_1 - {}^0\bar{K}_1 \mathbf{x}_1 + \sum_{k=1}^3 \left({}^0\hat{C}_{2k} \dot{\mathbf{x}}_{2p_{2k}-1} + {}^0\hat{K}_{2k} \mathbf{x}_{2p_{2k}-1} \right) \right] \end{aligned} \quad (\text{A.1.1})$$

$$\begin{aligned} \ddot{\mathbf{x}}_{24} = & \frac{1}{I_{z_{23}}} \left[\tau_{23}^0 - {}^0\bar{c}_{\theta,23} \dot{\mathbf{x}}_{24} - {}^0\bar{k}_{\theta,23} \mathbf{x}_{24} + {}^0\hat{c}_{\theta,33} \dot{\mathbf{x}}_{26} + {}^0\hat{k}_{\theta,33} \mathbf{x}_{26} \right] \\ & - \frac{1}{I_{z_0}} \left[\tau_1^0 - {}^0\bar{C}_1 \dot{\mathbf{x}}_2 - {}^0\bar{K}_1 \mathbf{x}_2 + \sum_{k=1}^3 \left({}^0\hat{c}_{\theta,2k} \dot{\mathbf{x}}_{2p_{2k}} + {}^0\hat{k}_{\theta,2k} \mathbf{x}_{2p_{2k}} \right) \right] \end{aligned} \quad (\text{A.1.2})$$

Intermediate body ($i = 3, p = 13$):

$$\begin{aligned} \ddot{\mathbf{x}}_{25} = & \frac{1}{\bar{m}_{33}} \left[u_{33}^0 - {}^0\bar{C}_{33} \dot{\mathbf{x}}_{25} - {}^0\bar{K}_{33} \mathbf{x}_{25} + {}^0\hat{C}_{(4)3} \dot{\mathbf{x}}_{27} + {}^0\hat{K}_{(4)3} \mathbf{x}_{27} \right] \\ & - \frac{1}{\bar{m}_{(2)3}} \left[u_{(2)3}^0 - {}^0\bar{C}_{(2)3} \dot{\mathbf{x}}_{23} - {}^0\bar{K}_{(2)3} \mathbf{x}_{23} + {}^0\hat{C}_{33} \dot{\mathbf{x}}_{25} + {}^0\hat{K}_{33} \mathbf{x}_{25} \right] \end{aligned} \quad (\text{A.1.3})$$

$$\begin{aligned} \ddot{\mathbf{x}}_{26} = & \frac{1}{I_{z_{33}}} \left[\tau_{33}^0 - {}^0\bar{c}_{\theta,33} \dot{\mathbf{x}}_{26} - {}^0\bar{k}_{\theta,33} \mathbf{x}_{26} + {}^0\hat{c}_{\theta,(4)3} \dot{\mathbf{x}}_{28} + {}^0\hat{k}_{\theta,(4)3} \mathbf{x}_{28} \right] \\ & - \frac{1}{I_{z_{(2)3}}} \left[\tau_{(2)3}^0 - {}^0\bar{c}_{\theta,(2)3} \dot{\mathbf{x}}_{24} - {}^0\bar{k}_{\theta,(2)3} \mathbf{x}_{24} + {}^0\hat{c}_{\theta,33} \dot{\mathbf{x}}_{26} + {}^0\hat{k}_{\theta,33} \mathbf{x}_{26} \right] \end{aligned} \quad (\text{A.1.4})$$

Intermediate body ($i = 4, p = 14$):

$$\begin{aligned}\ddot{\mathbf{x}}_{27} = & \frac{1}{\bar{m}_{43}} \left[u_{43}^0 - {}^0\bar{C}_{43} \dot{\mathbf{x}}_{27} - {}^0\bar{K}_{43} \mathbf{x}_{27} + {}^0\hat{C}_{(5)3} \dot{\mathbf{x}}_{29} + {}^0\hat{K}_{(5)3} \mathbf{x}_{29} \right] \\ & - \frac{1}{\bar{m}_{(3)3}} \left[u_{(3)3}^0 - {}^0\bar{C}_{(3)3} \dot{\mathbf{x}}_{25} - {}^0\bar{K}_{(3)3} \mathbf{x}_{25} + {}^0\hat{C}_{43} \dot{\mathbf{x}}_{27} + {}^0\hat{K}_{43} \mathbf{x}_{27} \right]\end{aligned}\quad (\text{A.1.5})$$

$$\begin{aligned}\ddot{\mathbf{x}}_{28} = & \frac{1}{I_{z_{43}}} \left[\tau_{43}^0 - {}^0\bar{c}_{\theta,43} \dot{\mathbf{x}}_{28} - {}^0\bar{k}_{\theta,43} \mathbf{x}_{28} + {}^0\hat{c}_{\theta,(5)3} \dot{\mathbf{x}}_{30} + {}^0\hat{k}_{\theta,(5)3} \mathbf{x}_{30} \right] \\ & - \frac{1}{I_{z_{(3)3}}} \left[\tau_{(3)3}^0 - {}^0\bar{c}_{\theta,(3)3} \dot{\mathbf{x}}_{26} - {}^0\bar{k}_{\theta,(3)3} \mathbf{x}_{26} + {}^0\hat{c}_{\theta,43} \dot{\mathbf{x}}_{28} + {}^0\hat{k}_{\theta,43} \mathbf{x}_{28} \right]\end{aligned}\quad (\text{A.1.6})$$

Intermediate body ($i = 5, p = 15$):

$$\begin{aligned}\ddot{\mathbf{x}}_{29} = & \frac{1}{\bar{m}_{53}} \left[u_{53}^0 - {}^0\bar{C}_{53} \dot{\mathbf{x}}_{29} - {}^0\bar{K}_{53} \mathbf{x}_{29} + {}^0\hat{C}_{(6)3} \dot{\mathbf{x}}_{31} + {}^0\hat{K}_{(6)3} \mathbf{x}_{31} \right] \\ & - \frac{1}{\bar{m}_{(4)3}} \left[u_{(4)3}^0 - {}^0\bar{C}_{(4)3} \dot{\mathbf{x}}_{27} - {}^0\bar{K}_{(4)3} \mathbf{x}_{27} + {}^0\hat{C}_{53} \dot{\mathbf{x}}_{29} + {}^0\hat{K}_{53} \mathbf{x}_{29} \right]\end{aligned}\quad (\text{A.1.7})$$

$$\begin{aligned}\ddot{\mathbf{x}}_{30} = & \frac{1}{I_{z_{53}}} \left[\tau_{53}^0 - {}^0\bar{c}_{\theta,53} \dot{\mathbf{x}}_{30} - {}^0\bar{k}_{\theta,53} \mathbf{x}_{30} + {}^0\hat{c}_{\theta,(6)3} \dot{\mathbf{x}}_{32} + {}^0\hat{k}_{\theta,(6)3} \mathbf{x}_{32} \right] \\ & - \frac{1}{I_{z_{(4)3}}} \left[\tau_{(4)3}^0 - {}^0\bar{c}_{\theta,(4)3} \dot{\mathbf{x}}_{28} - {}^0\bar{k}_{\theta,(4)3} \mathbf{x}_{28} + {}^0\hat{c}_{\theta,53} \dot{\mathbf{x}}_{30} + {}^0\hat{k}_{\theta,53} \mathbf{x}_{30} \right]\end{aligned}\quad (\text{A.1.8})$$

Intermediate body ($i = 6, p = 16$):

$$\begin{aligned}\ddot{\mathbf{x}}_{31} = & \frac{1}{\bar{m}_{63}} \left[u_{63}^0 - {}^0\bar{C}_{63} \dot{\mathbf{x}}_{31} - {}^0\bar{K}_{63} \mathbf{x}_{31} + {}^0\hat{C}_{(7)3} \dot{\mathbf{x}}_{33} + {}^0\hat{K}_{(7)3} \mathbf{x}_{33} \right] \\ & - \frac{1}{\bar{m}_{(5)3}} \left[u_{(5)3}^0 - {}^0\bar{C}_{(5)3} \dot{\mathbf{x}}_{29} - {}^0\bar{K}_{(5)3} \mathbf{x}_{29} + {}^0\hat{C}_{63} \dot{\mathbf{x}}_{31} + {}^0\hat{K}_{63} \mathbf{x}_{31} \right]\end{aligned}\quad (\text{A.1.9})$$

$$\begin{aligned}\ddot{\mathbf{x}}_{32} = & \frac{1}{I_{z_{63}}} \left[\tau_{63}^0 - {}^0\bar{c}_{\theta,63} \dot{\mathbf{x}}_{32} - {}^0\bar{k}_{\theta,63} \mathbf{x}_{32} + {}^0\hat{c}_{\theta,(7)3} \dot{\mathbf{x}}_{34} + {}^0\hat{k}_{\theta,(7)3} \mathbf{x}_{34} \right] \\ & - \frac{1}{I_{z_{(5)3}}} \left[\tau_{(5)3}^0 - {}^0\bar{c}_{\theta,(5)3} \dot{\mathbf{x}}_{30} - {}^0\bar{k}_{\theta,(5)3} \mathbf{x}_{30} + {}^0\hat{c}_{\theta,63} \dot{\mathbf{x}}_{32} + {}^0\hat{k}_{\theta,63} \mathbf{x}_{32} \right]\end{aligned}\quad (\text{A.1.10})$$

End-effector body ($i = 7, p = 17$):

$$\begin{aligned}\ddot{\mathbf{x}}_{33} = & \frac{1}{\bar{m}_{73}} \left[u_{73}^0 - {}^0\bar{C}_{73} \dot{\mathbf{x}}_{33} - {}^0\bar{K}_{73} \mathbf{x}_{33} \right] \\ & - \frac{1}{\bar{m}_{(6)3}} \left[u_{(6)3}^0 - {}^0\bar{C}_{(6)3} \dot{\mathbf{x}}_{31} - {}^0\bar{K}_{(6)3} \mathbf{x}_{31} + {}^0\hat{C}_{73} \dot{\mathbf{x}}_{33} + {}^0\hat{K}_{73} \mathbf{x}_{33} \right]\end{aligned}\quad (\text{A.1.11})$$

$$\begin{aligned}\ddot{\mathbf{x}}_{34} = & \frac{1}{I_{z_{73}}} \left[\tau_{73}^0 - {}^0\bar{c}_{\theta,73} \dot{\mathbf{x}}_{34} - {}^0\bar{k}_{\theta,73} \mathbf{x}_{34} \right] \\ & - \frac{1}{I_{z_{(6)3}}} \left[\tau_{(6)3}^0 - {}^0\bar{c}_{\theta,(6)3} \dot{\mathbf{x}}_{32} - {}^0\bar{k}_{\theta,(6)3} \mathbf{x}_{32} + {}^0\hat{c}_{\theta,73} \dot{\mathbf{x}}_{34} + {}^0\hat{k}_{\theta,73} \mathbf{x}_{34} \right]\end{aligned}\quad (\text{A.1.12})$$